

POWER ENHANCEMENT WITH GRID STABILIZATION OF RENEWABLE ENERGY-BASED GENERATION SYSTEM USING UPQC-FLC-EVA TECHNIQUE

A Major project report submitted in partial fulfilment of the degree of the
Bachelor of Technology in Electrical and Electronics Engineering

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CERTIFICATE



This is to certify that the Major Project Report entitled “**POWER ENHANCEMENT WITH GRID STABILIZATION OF RENEWABLE ENERGY BASED GENERATION SYSTEM USING UPQC FLC EVA TECHNIQUE**” is a Bonafide work the students **K.Epsiba(22C45A0215),B.Ashadevi(22C45A0206),MD.Sohail(22C45A0248), Ch.Karthik(22C45A0215)** submitted in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology** in **Electrical and Electronics Engineering** during the academic year **2024-2025**

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ACKNOWLEDGMENT

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ABSTRACT

The proposed work focuses on the power enhancement of grid-connected solar photovoltaic and wind energy PV-WE system integrated with an energy storage system ESS and electric vehicles EVs. The research works available in the literature emphasize only on PV, PV-ESS, WE, and WE-ESS. The enhancement techniques such as Unified Power Flow Controller UPFC, Generalized UPFC GUPFC, and Static Var Compensator SVC and Artificial Intelligence AI-based techniques including Fuzzy Logic Controller FLC-UPFC, and Unified Power Quality Conditioner UPQC-FLC have been perceived in the existing literature for power enhancement. Further, the EVs are emerging as an integral domain of the power grid but because of the uncertainties and limitations involved in renewable energy sources RESs and ESS, the EVs preference towards the RES is shifted away. Therefore, it is required to focus on improving the power quality of the PV-WE-ESS-EV system connected with the grid, which is yet to be explored and validated with the available technique for enhancing power quality. Furthermore, in the case of the bidirectional power flow from vehicle-to-grid V2G and grid-to-vehicle G2V, optimal controlling is crucial for which an electric vehicle aggregator EVA is designed. The designed EVA is proposed for the PV-WE-ESS-EV system so as to obtain the benefits such as uninterruptible power supply, effective the load demand satisfaction, and efficient utilization of the electrical power. The power flow from source to load and from one source to another source is controlled with the support of FLC.

The FLC decides the economic utilization of power during peak load and off-peak load. The reduced power quality at the load side is observed as a result of varying loads in the random fashion and this issue is sorted out by using UPQC in this proposed study. From the results, it can be observed that the maximum power is achieved in the case of PV and WE systems with the help of the FLC-based maximum power point tracking MPPT technique. Furthermore, the artificial neural network ANN-based technique is utilized for the development of the MPPT algorithm which in turn is employed for the validation of the proposed technique. The outputs of both the techniques are compared to select the best-performing technique. A key observation from the results and analysis indicates that the power output from FLC-based MPPT is better than that of ANN-based MPPT. Thus, the proper and economical utilization of power is achieved with the help of FLC and UPQC. It can be inferred that the EVs can play a vital role in imparting the flexibility in terms of power consumption and grid stabilization during peak load and off-peak load durations provided that the proper control techniques and grid integration are well-established.

Chapter-1

INTRODUCTION

The conventional energy resources are being substituted by RESs, such as solar, wind, tidal and many others. Also, the utilization of RESs is increasing tremendously because these are clean, easy to access, and abundant in nature. Moreover, a favourable scenario such as the reduction in global warming and less emission of carbon dioxide has been observed when RESs are implemented. Therefore, RESs prevail as better alternatives to conventional resources. The increasing load demand can also be fulfilled by introducing RESs in existing power grid [1]. In spite of having numerous advantages, the solar and wind energy resources are highly sensitive to the weather and geographical location, which are the prime drawbacks of using RESs. These drawbacks affect the grid stability to the greater extent which can be potentially minimized by integrating the RESs with ESS.

The various problems associated with RESs and the process to overcome those problems are detailed in the referred study. The uncertainties involved in PV system cause Total Harmonic Distortion THD in the grid. As a result of which the load voltage and current are distorted. Therefore, the Flexible AC Transmission System FACTS devices have been utilized to overcome the THD and voltage stabilization issues. The FACTS devices, including UPFC, UPFCFLC, GUPFC, UPQC, and SVC are the commonly used devices to improve power quality. The implementation of FACTS devices resolves the voltage stability issue and simultaneously, improves the grid power quality. The enhancement in grid stability and reduction in THD have been achieved by implementing the Static Compensator (STATCOM) accompanying with ESS. Similarly, wind power generation system experiences unstable voltage and THD. Therefore, STATCOM has also been implemented with the grid-connected WE system [6]. The three FACTS devices, including Dynamic Voltage Restoration (DVR), Distribution STATCOM (DSTATCOM), and UPQC have been implemented for power quality improvement and voltage stabilization. Moreover, the implementation of UPQC in the grid-connected PV system has been discussed for power quality enhancement. The WE integrated grid system has been considered for power quality improvement using UPQC. The power quality of the grid connected RESs can be improved using various MPPT algorithms based on FLC and ANN. Further, the uncertainties involved in RESs can be avoided using FLC-based

MPPT algorithms. The FLC-based MPPT have been implemented for avoiding the weather induced uncertainties related to the solar PV output power . The authors concluded that the proposed method has overcome the conventional methods for power quality enhancement.

The converters are used along with the RESs to convert the power and make it suitable for connecting to the grid. The configuration layout of hybrid microgrid, including PV and WE has been provided in the referred study in which the RESs are connected with a bidirectional converter. Various factors including average wind speed, irradiation, and temperature are needed to be considered while modelling the PV and WE systems as RESs are sensitive to weather and location. The scenario of RESs for various countries such as India, China, Iceland, Sweden, and the USA differ widely . To add upon it in WE system, the output power capacity is different on onshore and offshore. Therefore, depending upon the various factors affecting the PV and WE systems, flexible mathematical model and optimization techniques are required to model the seasonable WE and PV systems. The non-linear characteristics of PV are challenging to design. Therefore, various optimization techniques like particle swarm optimization are used to estimate the parameters associated with the PV system and to improve the power output of PV and WE systems . These techniques would ultimately help in configuring the RESs modelling. The RESs configurations available in literature are illustrated in Table 1.

Among the power electronic-based FACTS techniques, the UPQC has better performance when compared to other techniques, essentially STATCOM, DVR, and SVC. The UPQC fulfills most of the desired qualities, including voltage improvement, harmonic mitigation, current improvement, and overall power quality improvement of the system , . Therefore, UPQC is utilized in this study to improve the power output quality of the PV-WE-ESS-EV system integrated with EVs and ESS. A dynamic battery controller is designed to effectively control the ESS system. On the other hand, EVA is implemented for controlling the EVs system, which manages the power transfer between the PVWE- ESS system and EVs. The UPQC connected with the PV-WE-ESS-EV system functions in two ways which are as follows

- The series part of UPQC (active converter) helps to reduce the harmonic mitigation and voltage disturbances.
- The shunt part of UPQC (active converter) helps to reduce the current distortion and improve the dc-link voltage regulation.

The implementation of UPQC for a solar PV system and its investigation under an unbalanced loading condition has also been carried out [29]. The authors concluded that the implemented UPQC was able to improve both the power quality and harmonic mitigation. The controlling of the PV-UPQC system has been done using a proportional-integral (PI) controller, FLC, or neural network (NN)-based algorithm. The PI controller has been used to control the working of PV-UPQC [30] and FLC technique has been used for improving the power quality in the PV-UPQC system integrated with a battery storage system (BSS) [27]. The authors have provided a comparative analysis between the PI and FLC techniques and concluded that the FLC technique gives better results for the PV-UPQC-BSS system relative to the other technique based on PI. The implementation of the NN approach in PV-UPQC for power quality improvement has been mentioned [31], [32]. This technique requires proper learning and validation before implementing it to the electrical system. It consumes more time to evaluate the output response of the system but it gives accurate and appropriate responses. EVs also support to improve the power quality when used in parallel with grid connected ESS integrated RESs. The role of EVs, charging schemes, and benefits are discussed in [33]–[35]. The EVs are mainly for transportation purpose and thus, the availability of EVs is uncertain. For this issue, the EVA is used to manage the availability and to allot the charging-discharging schedule of EVs connected to the grid [36], [37]. The EVA enables the grid system to control the charging and discharging operation and acts as a mediator controller between the grid and the vehicles. This helps the grid system to improve the quality of power, reduce the THD, load balancing, and to improve the voltage profile under different load unbalancing conditions. Several controlling techniques and optimization techniques of the EV system connected with the grid have been discussed

Chapter-2

LITERATURE SURVEY

INTRODUCTION:

This project introduces a new topology, yet simple and efficient, for a grid-connected wind-solar cogeneration system. A permanent magnet synchronous generator-based full-scale wind turbine is interconnected to the utility-grid via back-to-back voltage-source converters (VSCs). The dc-link capacitor has been utilized to directly interface a photovoltaic solar generator. No dc/dc conversion stages are required, and hence, the hybrid system is simple and efficient.

EXISTING SYSTEM:

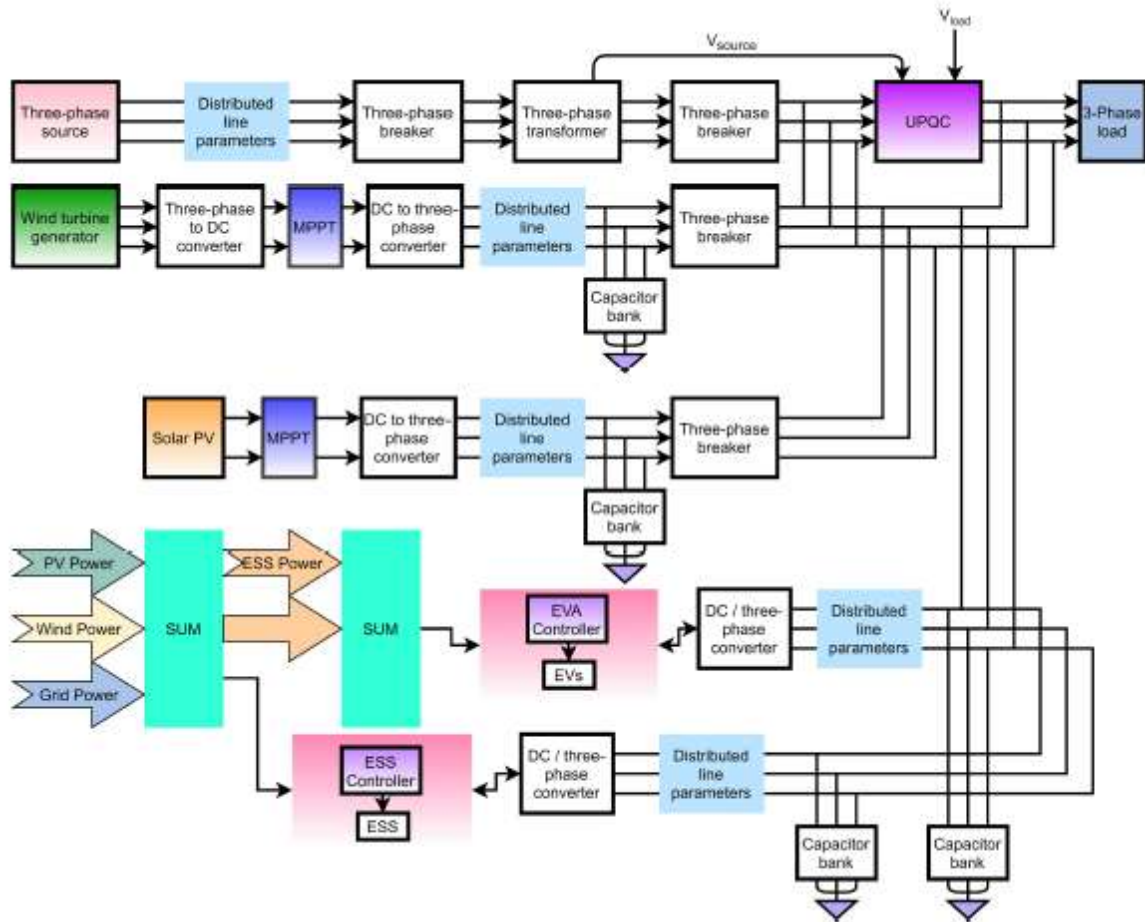
In existing system the power electronic-based FACTS techniques are operated with help of ANN controller, the UPQC has better performance when compared to other techniques, essentially STATCOM, DVR, and SVC. The UPQC fulfills most of the desired qualities, including voltage improvement, harmonic mitigation, current improvement, and overall power quality improvement of the system. Therefore, UPQC is utilized in this study to improve the power output quality of the PV-WE-ESS-EV system integrated with EVs and ESS. A dynamic battery controller is designed to effectively control the ESS system. On the other hand, EVA is implemented for controlling the EVs system, which manages the power transfer between the PVWE- ESS system and EVs.

PROPOSED SYSTEM:

The proposed work focuses on the power enhancement of grid-connected solar photovoltaic and wind energy (PV-WE) system integrated with an energy storage system (ESS) and electric vehicles (EVs). The research works available in the literature emphasize only on PV, PV-ESS, WE, and WE-ESS. The enhancement techniques such as Unified Power Flow Controller (UPFC), Generalized UPFC (GUPFC), and Static Var Compensator (SVC) and Artificial Intelligence (AI)-based techniques including Fuzzy Logic Controller (FLC)-UPFC, and Unified Power Quality Conditioner (UPQC)-FLC have been perceived in the existing literature for power enhancement. Further, the EVs are emerging as an integral domain of the power grid but because of the uncertainties and limitations involved in renewable energy sources (RESs) and ESS, the EVs preference towards the RES is shifted away. Therefore, it is required to focus on improving the power quality of the PV-WE-ESS-EV system connected with the grid, which

is yet to be explored and validated with the available technique for enhancing power quality. Furthermore, in the case of the bidirectional power flow from vehicle-to-grid (V2G) and grid-to-vehicle (G2V), optimal controlling is crucial for which an electric vehicle aggregator (EVA) is designed. The designed EVA is proposed for the PV-WE-ESS-EV system so as to obtain the benefits such as uninterruptible power supply, effective the load demand satisfaction, and efficient utilization of the electrical power. The power flow from source to load and from one source to another source is controlled with the support of FLC. The FLC decides the economic utilization of power during peak load and off-peak load. The reduced power quality at the load side is observed as a result of varying loads in the random fashion and this issue is sorted out by using UPQC in this proposed study. From the results, it can be observed that the maximum power is achieved in the case of PV and WE systems with the help of the FLC-based maximum power point tracking (MPPT) technique. Furthermore, the artificial neural network (ANN)-based technique is utilized for the development of the MPPT algorithm which in turn is employed for the validation of the proposed technique.

BLOCK DIAGRAM:



Proposed diagram.

Chapter-3

WIND POWER GENERATION

Wind power is making use of air circulation with wind generators to supply the mechanical power to transform electrical generators. Wind power, as an alternate to shedding nonrenewable fuel sources, abounds, sustainable, extensively dispersed, tidy, creates no gas discharges throughout procedure, takes in no water, as well as utilizes little land. The internet impacts on the setting are much less troublesome compared to those of nonrenewable source of power. Wind ranches contain numerous specific wind generators, which are linked to the electrical power transmission network. Onshore wind is a low-cost resource of electrical power, affordable with or in lots of locations less costly compared to coal or gas plants. Offshore wind is steadier and also more powerful compared to ashore as well as overseas branches have much less aesthetic effect, however building as well as upkeep expenses is substantially greater. Little onshore wind ranches could feed some power right into the grid or supply electrical power to separated off-grid areas.

Wind power offers variable power, which is extremely regular from year to year however has considerable variant over much shorter time ranges. It is for that reason utilized combined with various other electrical source of power to provide a trusted supply. As the percentage of wind power in an area boosts, a should update the grid as well as a decreased capability to replace traditional manufacturing could take place. [7] [8] Power-management methods such as having excess capability, geographically dispersed wind turbines, send off able support resources, adequate hydroelectric power, exporting and also importing power to bordering locations, or decreasing need when wind manufacturing is reduced, could in most cases get rid of these troubles. On top of that, weather condition projecting allows the electric-power network to be prepped for the foreseeable variants in manufacturing that happen. Since 2015, Denmark produces 40% of its electrical power from wind, as well as at the very least 83 various other nations around the globe are utilizing wind power to provide their electrical power grids. In 2014, international wind power capability broadened 16% to 369,553 MW. Annual wind power manufacturing is additionally proliferating and also has actually gotten to around 4% of globally electrical power use, 11.4% in the EU.

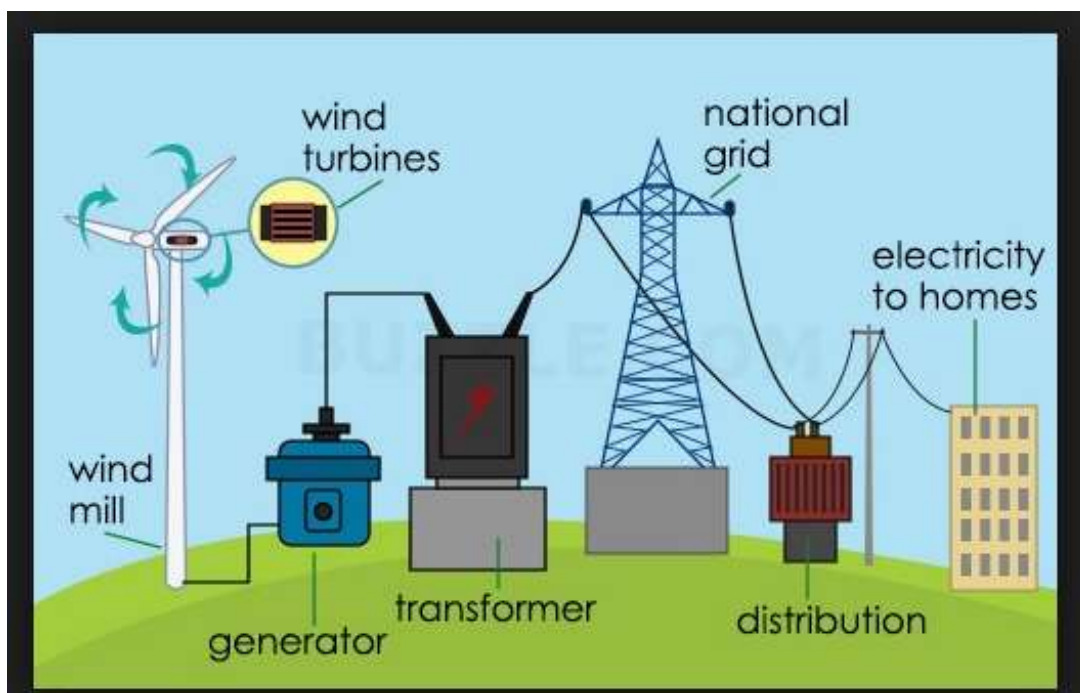
WIND FRAMES:

A wind ranch is a team of wind generators in the very same area utilized for manufacturing of electrical power. A big wind ranch might contain a number of hundred specific wind generators dispersed over an extensive location, yet the land in between the wind turbines could be utilized for farming or various other functions. As an example, Gansu Wind Farm, the biggest wind ranch on the planet, has numerous thousand generators. A wind ranch might additionally lie offshore. Nearly all big wind generators have the exact same style a straight axis wind generator having an upwind blades with 3 blades, connected to a nacelle in addition to a high tubular tower. In a wind ranch, specific wind turbines are adjoined with a tool voltage (usually 34.5 kV), power collection system as well as interactions network. Generally, a range of $7D$ ($7 \times$ Rotor Diameter of the Wind Turbine) is established in between each generator in a totally created wind ranch. At a substation, this medium-voltage electrical current is raised in voltage with a transformer for link to the high voltage electrical power transmission system. Induction generators, which were frequently utilized for wind power tasks in the 1980s as well as 1990s, need responsive power for excitation so substations utilized in wind-power collection systems consist of considerable capacitor financial institutions for power aspect adjustment. Various kinds of wind generator generators act in different ways throughout transmission grid disruptions, so considerable modeling of the vibrant electromechanical attributes of a brand-new wind ranch is needed by transmission system drivers to guarantee foreseeable secure actions throughout system mistakes. Specifically, induction generators could not sustain the system voltage throughout mistakes, unlike vapor or hydro turbine-driven concurrent generators. Today these generators typically aren't made use of anymore in modern-day generators. Rather today most wind turbines utilize variable rate generators incorporated with partial- or major power converter in between the wind turbine generator as well as the collection agency system, which typically have better homes for grid affiliation and also have Low voltage flight through-capabilities. [50] Modern ideas make use of either twice as fed devices with partial-scale converters or squirrel-cage induction generators or simultaneous generators (both completely and also electrically delighted) with complete range converters. Transmission systems drivers will provide a wind ranch programmer with a grid code to define the needs for affiliation to the transmission grid. This will certainly consist of power element, regularity of regularity as well as vibrant actions of the wind ranch generators throughout a system mistake.

OFFSHORE WIND POWER:

Offshore wind power describes the building and construction of wind ranches in big bodies of water to create electrical power. These setups could use the extra constant and also effective winds that are offered in these areas as well as have much less visual effect on the landscape compared to land based jobs. Nevertheless, the building and also the upkeep prices are significantly greater. Siemens and also Vestas are the leading wind turbine vendors for overseas wind power. Ørsted, Vattenfall and also E.ON are the leading overseas drivers. [56] Since October 2010, 3.16 GW of overseas wind power ability was functional, primarily in Northern Europe. Inning Accordance With BTM Consult, greater than 16 GW of extra ability will certainly be mounted prior to completion of 2014 as well as the UK and also Germany will certainly end up being both leading markets. Offshore wind power ability is anticipated to get to a total amount of 75 GW globally by 2020, with considerable payments from China and also the United States. [56] The UK's financial investments in overseas wind power have actually led to a quick decline of the use of coal as a power resource in between 2012 and also 2017, in addition to a decrease in the use of gas as a power resource in 2017.

In 2012, 1,662 generators at 55 overseas wind ranches in 10 European nations created 18 TWh, sufficient to power virtually 5 million homes. Since August 2013 the London Array in the United Kingdom is the biggest overseas wind ranch worldwide at 630MW.



Wind power generation system.

COLLECTION AND TRANSMISSION NETWORK:

Among the most significant present difficulties to wind power grid combination in the United States is the requirement of establishing brand-new transmission lines to bring power from wind ranches, normally in remote lowly booming states in the center of the nation as a result of accessibility of wind, to high lots areas, generally on the coastlines where populace thickness is greater. The present transmission lines in remote areas were not developed for the transportation of huge quantities of power. As transmission lines end up being much longer the losses related to power transmission boost, as settings of losses at reduced sizes are exacerbated as well as brand-new settings of losses are not minimal as the size is boosted, making it tougher to transfer huge lots over big ranges. [68] Nevertheless, resistance from state as well as city governments makes it challenging to create brand-new transmission lines. Multi state power transmission tasks are dissuaded by states with inexpensive electrical power prices for concern that exporting their economical power will certainly cause enhanced prices. A 2005 power legislation provided the Energy Department authority to authorize transmission jobs states chose not to act upon, however after an effort to utilize this authority, the Senate stated the division was being extremely hostile in doing so. One more issue is that wind firms discover after that the bandwidth of a brand-new ranch is listed below the generation capability, mainly due to the fact that government energy policies to urge renewable resource setup enable feeder lines to fulfill just minimal requirements. These are necessary concerns that should be fixed, as when the bandwidth does not fulfill the generation capability, wind ranches are required to create listed below their complete capacity or quit running completely, in a procedure referred to as curtailment. While this brings about possible eco-friendly generation left untapped, it avoids feasible grid overload or threat to dependable solution.



Offshore wind power.

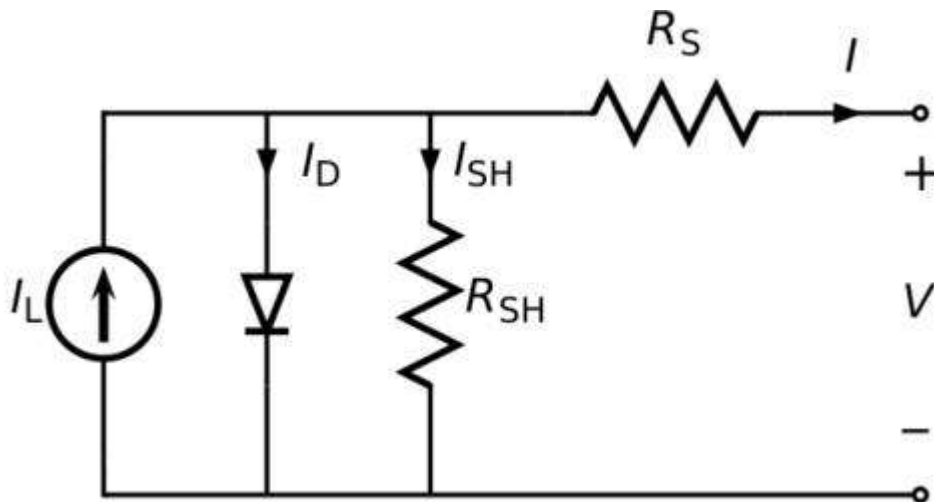
SOLAR POWER GENERATION:

Introduction:

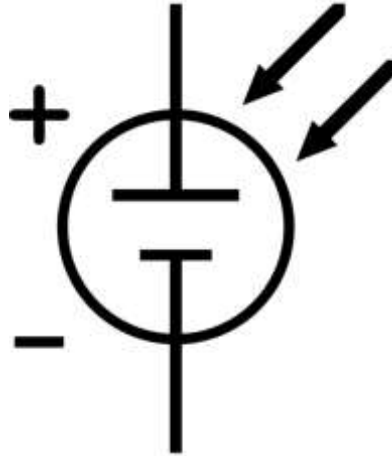
A solar cell or photovoltaic cell is a device that converts solar energy into electricity by the photovoltaic effect. Sometimes the term solar cell is reserved for devices intended specifically to capture energy from sunlight, while the term photovoltaic cell is used when the source is unspecified. Assemblies of cells are used to make solar panel, solar modules, or photovoltaic arrays. Photovoltaic is the field of technology and research related to the application of solar cells for solar energy. Solar cell efficiencies vary from 6% for amorphous silicon-based solar cells to 40.7% with multiple-junction research lab cells and 42.8% with multiple dies assembled into a hybrid package. Solar cell energy conversion efficiencies for commercially available multicrystalline Si solar cells are around 14-19%.

Solar cells can also be applied to other electronics devices to make it self-power sustainable in the sun. There are solar cell phone chargers, solar bike light and solar camping lanterns that people can adopt for daily use

Equivalent circuit of a solar cell



The equivalent circuit of a solar cell



The schematic symbol of a solar cell

1. Photons in sunlight hit the solar panel and are absorbed by semi conducting materials, such as silicon.
2. Electrons (negatively charged) are knocked loose from their atoms, allowing them to flow through the material to produce electricity. Due to the special composition of solar cells, only allow the electrons to move in a single direction. The complementary positive charges that are also created (like bubbles) are called holes and flow in the direction opposite of the electrons in a silicon solar panel.
3. An array of solar panels converts solar energy into a usable amount of direct current (DC) electricity.

Solar battery chargers are better for the environment in a few ways. For one, with them, batteries can be recharged, therefore no longer contributing to growing landfills. Also, batteries have potentially harmful metals inside them – we do not want to be simply throwing them out into landfills!

Also, if using batteries that can be recharged with a solar battery charger, a person can stop wasting his or her money on the purchase of new batteries.

The batteries of cell phones, PDAs, laptops, mp3 players, and more can be charged by solar battery chargers. This means that you do not have to rely on electricity to charge these devices.

This is especially good because most electricity is created by non-sustainable, polluting methods.

Solar battery chargers are also good because they allow the users to charge devices, even when no power outlets are around. This makes them especially useful when working out in the field, traveling, hiking, and/or during an emergency.

It does not matter if you use a 12 volt solar battery charger, a solar car battery charger, or any other type of charger – the use of any solar battery charger is going to help alleviate waste. By supporting and using solar battery chargers and other sustainable technology, you can help make a difference in today's world.

The solar cells positive terminal is connected through the diode to the positive terminal of the 1.2V battery. If the voltage of the solar cell drops below 1.4 volts then with the 0.2V the blocking diode takes there wont be enough potential to charge the 1.2V battery. The purpose of the diode is to disallow current dissipating out from the battery to the solar cell when this low voltage situation occurs in the solar cell.



A solar battery is one of the most important energy sources available to save energy consumption, and serves as a spare source while normal power supply shuts down. Systems using solar batteries have various scales from a few watts to a few thousands of kilowatts, and also have various types. Conventionally, the solar battery has been dominantly used in the form of a solar electricity generation plant where a large number of solar batteries are arranged, or used for securing power supply at a remote location. Recently, it becomes more and more popular to install a solar battery module panel on a house roof or on an outer wall of a building. Generally, a solar battery is composed of a plurality of photoelectric power generating elements connected in series on a substrate to obtain a photo voltage. In a solar panel battery, the solar cell is the smallest constituent unit of a device having the function of photoelectric conversion. The solar cell is considered a major candidate for obtaining energy from sun, since it can convert sunlight directly to electricity with high conversion efficiency, can provide nearly permanent power at low operating cost without having any influence on the climate.

Working:

Solar power systems employ photovoltaic cells to convert the radiant energy of sunlight directly into electrical energy. Photovoltaic solar cells are semiconductor devices which convert sunlight into electricity. Solar cells which utilize crystalline semiconductors, such as silicon, offer the advantages of high performance and reliability. Photovoltaic cells are silicon-base crystal wafers which produce a voltage between opposite surfaces when light strikes one of the surfaces, which surface has a current collecting grid thereon. The photons of the light are absorbed by photovoltaic cells and yield their energy to the valence electrons of the semiconductor and tear them from the bonds that maintain them joined to the cores of the atoms, promoting them to a superior energetic state called conduction band in which they can move easily through the semiconductor.

Typically, a plurality of solar cells are assembled and interconnected so as to form a physically-integrated module, and then a number of such modules are assembled together to form a solar panel. Several solar panels may be connected together to form a larger array. The individual photovoltaic cells in a module may be connected in series or parallel, typically by an internal wiring arrangement and similarly two or more modules in a panel may be connected in series or parallel, depending upon the voltage output desired. Solar cells are usually interconnected into series strips by electrically interconnecting a collector pad on the grid to the opposite

surface of the adjacent cell in the strip. Photovoltaic cells are manufactured in a variety of configurations, but generally comprise a layered structure on a substrate. There are many different types of converging solar cell modules in which sunlight is converged by means of a lens system so that the total area of expensive solar cells can be reduced in order to reduce the cost of electric power generating systems using these solar cells. In order to most efficiently use the electrical power generated by a photovoltaic cell or photovoltaic array, it is desirable to maximize the power generated by the photovoltaic cell or photovoltaic array, despite varying weather conditions. Various sun tracking systems have been used to enhance the power generating efficiency of the converging solar cell module.

Theory:

A solar energy battery is different from the regular battery. The solar battery module is constructed by having a multiplicity of solar battery elements carried on a supporting base plate. When the sunlight impinges on the individual solar battery elements, the energy of the light which makes no contribution to the photoelectric conversion is accumulated in the form of heat to elevate the temperature of the solar battery elements and lower the efficiency of photoelectric conversion. A solar cell having a photoelectric conversion layer in which at least one PIN junction is formed using an amorphous or microcrystalline silicon film is utilized. A solar battery converts light into electrical energy, its P-N junction structure when exposed to incident light generates large quantities of electron-hole pairs, and in the meantime electrons carrying negative electricity and holes carrying positive electricity migrate to the N-type semiconductor and P-type semiconductor respectively. This process produces electricity. In such converging solar cell modules, converging solar cell elements each having solar cells and their electrodes for outputting electric currents are used. When a spot formed by converged sunlight irradiates the light receiving surface of the converging solar cell, free electrons and electron holes as carriers are generated inside a silicon substrate. The photoelectric conversion efficiency of a solar battery depends mainly on the internal resistance of the solar battery. In particular, it depends on the series resistance of the upper and lower electrodes, and the series resistance of the elongation of the upper and lower electrodes which are brought into contact with each other in order to connect adjacent generating regions.

A typical solar battery comprises a glass substrate as a front side transparent protective member at a light-receiving side, a back side protective member, ethylene-vinyl acetate copolymer

(EVA) films as sealing films arranged between the glass substrate and the back side protective member, and solar cells or silicon photovoltaic elements sealed by the EVA films. A solar battery module is generally composed of a solar battery panel comprising a light-transmission panel and a solar battery element, the solar battery element being provided on the surface which is opposed to the light-receiving surface of the light-transmission plate, and a frame for fixing the solar battery panel thereon. Solar panels have a large number of solar cells which are used to convert power from sunlight.

Power generated by the solar cells is coupled via electric lines to a rectifier for feeding into the alternating current (AC) network or to a battery. A connecting box is generally provided for coupling to the solar panel. Solar panels are comprised primarily of a strong back, insulation, receiver tubes, headers and tube guide/supports. Tubes are connected at the top and bottom of the panel by the headers. Solar panels are typically mounted on a mounting structure, which is supported on a mounting surface, such as a rooftop. The sun's thermal energy is intercepted by a collector system that is comprised of thousands of sun tracking mirrors called heliostats. This energy is redirected and concentrated on a heat exchanger, called a solar receiver. The receiver includes a plurality of solar receiver panels positioned around an outside wall of the receiver. A solar battery module panel has a plurality of photovoltaic elements resin-sealed between a surface cover glass and a back cover film. In the case where several modules are to be interconnected, and also in the case where two or more solar panels are to be interconnected, external terminals are required for connections to cables that couple the modules or panels together.

Solar batteries have been used in various electronic equipments as power supply substitutes for dry batteries. Such batteries are highly reliable, have a long life, and now are economically produced. Initially, large-surfaced solar cell arrangements are used in photo-voltaic systems, for example, which can provide sufficient energy for consumers with a higher demand. Solar panels are particularly well suited to situations where electrical power from the grid is unavailable, such as in remote area power systems. Low power consumption electronic equipment such as electronic desktop calculators, watches, and portable electronic equipment

(e.g., digital cameras, cellular phones and commercial radar detectors) can be fully driven by the electromotive force of solar batteries.

Applications: ☒

Solar cells can also be applied to other electronics devices to make it self-power sustainable in the sun. There are solar cell phone chargers, solar bike light and solar camping lanterns that people can adopt for daily use.

Solar power plants can face high installation costs, although this has been decreasing due to the learning curve. Developing countries have started to build solar power plants, replacing other sources of energy generation.

In 2008, solar power supplied 0.02% of the world's total energy supply. Use has been doubling every two, or fewer, years. If it continued at that rate, solar power would become the dominant energy source within a few decades.

Since solar radiation is intermittent, solar power generation is combined either with storage or other energy sources to provide continuous power, although for small distributed producer/consumers, net metering makes this transparent to the consumer.

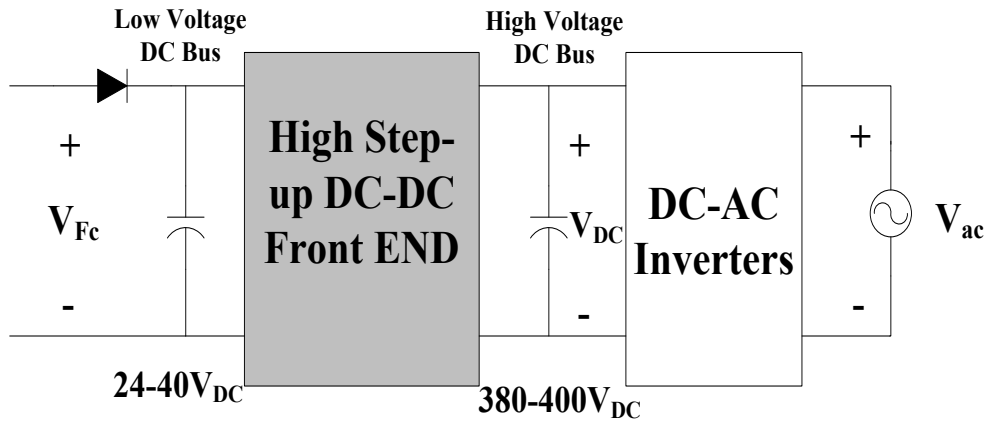
CHAPTER-4

DC TO DC CONVERTER

Introduction:

A DC– DC Converter with a high development up voltage gain is used for a few applications, for instance, high-drive discharge light adjusts for Automobile headlamps, Fuel Cell Energy Conversion structures, Solar Cell Energy Conversion systems and Battery support systems for Uninterruptible Power Supplies. Speculatively, a DC– DC Boost Converter can achieve a high development up voltage gain with an extraordinarily high commitment extent. In any case, basically, the movement up voltage gain is confined as a result of the effect of power switches, rectifier diodes and the indistinguishable game plan restriction of inductors and capacitors. At the point when all is said in done, a Conventional Boost Converter can be grasped to surrender a high development voltage gain with a sweeping commitment extent. Nevertheless, the Conversion efficiency and the movement up voltage gain are limited as a result of the necessities of the mishaps of impact switches and diodes, the proportionate game plan resistance of inductors and capacitors and the pivot recovery issue of diodes. In any case, the dynamic switch of these converters will bear high voltage stress and high power dispersal due to the spillage inductance of the transformer. To decrease the voltage spike, a resistor– capacitor– diode snubber can be used to restrict the voltage weight on the dynamic switch. Nevertheless, the efficiency will be reduced. High development up converters with a low data current swell in perspective of the coupled inductor has been made. The low data current swell of these converters is recognized by using an additional LC circuit with a coupled inductor.

Regardless, spillage inductance issues that relate to the Voltage Spike and capability remain colossal. A planned Boost– fly back converter in perspective of a coupled inductor with high viability and high development up voltage gain has been presented. The imperativeness set away in the spillage inductor is reused into the yield in the midst of the kill time frame. Thusly, the capability can be extended and the voltage weight on the dynamic switch can be smothered. Many developments up converters, which use a yield voltage stacking to grow the voltage gain, are shown.



General Power generation system with a high step-up converter

A high development up DC– DC Converter is showed up in Figure 3.1 with a joined coupled inductor and a regular mode electromagnetic block diminish channel. Here a Step up-fly back converter with a coupled inductor and a yield voltage stacking is made. A high development up converter, which utilizes a coupled inductor and a voltage doubler methodology on the yield voltage stacking to achieve a high development up voltage gain, is exhibited. A high development up help converter that uses various coupled inductors for the yield voltage stacking is proposed.

In addition, adventure up converters, which use a voltage lift, are displayed. Since the switch must persevere through high current in the midst of the switch on period, this framework is fitting for low-yield control applications. Since the low voltage rating and the low driving restriction $R_{DS(on)}$ of the power switch are used for these converters, the high change capability can be refined. Regardless, the need for a coupled inductor with a high coupling coefficient will realize amassing inconvenience and cost increment. A high development up converter, which uses a three state trading cell and a voltage multiplier mastermind in perspective of capacitors, can achieve high development up gain.

The Power outlining is the system used to supply electrical imperativeness from a source to its customers. It is of basic noteworthiness to industry. It is likely that the air we breathe in and water we drink are thought little of until the point that they are not there. Imperativeness change procedure is the central point of convergence of force building.

The looking at equipment can be detached into four social affairs:

- AC/AC transformers
- AC/DC rectifiers
- DC/DC converter

The Grid interconnection of PV/Fuel Cell structure requires control converters to meet the system essentials like Voltage adequacy, repeat and stage point. First follower the low voltage dc into high voltage dc by using bolster dc-dc converter and a short time later proselyte this dc voltage into cooling by using inverters in conclusion interface the whole system to grid. This kind of system i.e., dc-dc and dc-cooling change is called two stage change structure.

The DC-DC converters are electronic devices used at whatever point we have to change DC electrical power capably beginning with one voltage level then onto the following. They are required in light of the way that not in any way likes AC; DC can't simply be wandered up or down using a transformer. From numerous perspectives, a DC-DC converter is the thing that should be known as a transformer.

The dc-dc converters can be viewed as dc transformer that passes on a dc voltage or current at a surprising level in contrast with the data source. Electronic trading plays out this dc change as in customary transformers and not by electromagnetic means. The dc-dc converters find wide applications in coordinated switch-mode dc control supplies and in dc motor drive applications.

The DC-DC converters are non-coordinate in nature. The framework of first class control for them is a test for both the control building authorities and power devices engineers. At the point when all is said in done, an extraordinary control for dc-dc converter reliably ensures soundness in optional working condition. What's more, extraordinary response to the extent expulsion of load assortments, input voltage changes and altogether parameter vulnerabilities is furthermore required for a typical control plot.

After pioneer examination of dc-dc converters, a great deal of undertakings has been facilitated in working up the exhibiting and control systems of various dc-dc converters. Incredible direct approach relies upon the state averaging frameworks to get the state-space touched base at the midpoint of conditions. From the state-space landed at the midpoint of model, possible disturbances are brought into the state factors around the working point. In light of the conditions, trade components of the open-circle plant can be gained. A straight controller is definitely not hard to be illustrated with these basic trade limits in perspective of the trade work.

The DC to DC converters are indispensable in helpful electronic contraptions, for instance, cell phones and PCs, are given control from batteries in a general sense. Such electronic devices consistently contain a couple of sub-circuits, each with its own particular voltage level essential not the same as that given by the battery or an outside supply i.e., every so often higher or lower than the supply voltage and possibly even negative voltage. Besides, the battery voltage diminishes as its set away power is exhausted. Changed DC to DC converters offer a methodology to extend voltage from a to some extent cut down battery voltage thusly saving space instead of using distinctive batteries to accomplish a comparative thing.

DC-DC converters are electronic devices that are used at whatever point we have to change DC electrical power profitably beginning with one voltage level then onto the following. Insipidly the use of a switch or switches with the ultimate objective of power change can be seen as a SMPS. From now onwards at whatever point we say DC-DC Converters we will convey them with respect to SMPS.

A few usages of energy of DC-DC converters are the place 5V DC on a PC motherboard must be wandered down to 3V, 2V or less for a standout amongst the latest CPU chips; where 1.5V from a lone cell must be wandered up to something like 5v, to work electronic equipment. In these applications, we have to change the DC imperativeness beginning with one voltage level then onto the following, while at the same time misusing as pitiful as possible at the same time. By the day's end, we have to play out the change with the most raised possible capability.

Types of DC-DC Converters:

There are an extensive variety of sorts of DC-DC converters, each one of which tends to be more sensible for some sort of uses than for other individuals. For comfort they can be requested into various social events. Regardless, for example a couple of converters are fitting for wandering down the voltage, while others are sensible for wandering it up and a third assembling can be used for either. Here we will see major sorts of DC-DC converters.

At this moment DC-DC converters can be confined into two sorts as takes after.

- Non-limited dc-dc converters
- Isolated dc-dc converters

Non-limited DC-DC Converters:

The non-limited converter typically uses an inductor and there is no dc voltage separation between the information and the yield. Most by a long shot of usages don't require dc segregation between its information and yield voltages. The non-restricted dc-dc converter has a dc route between its data and yield. Battery-based systems that don't use the climate control system control line address an imperative application for non-isolated dc-dc converters. Reason for stack dc-dc converters that draw input control from a limited dc-dc converter, for instance, a vehicle converter address another comprehensively used non-isolated application.

Most of these dc-dc converter ICs use either an internal or external synchronous rectifier. Their single alluring fragment is ordinarily a yield inductor and along these lines less powerless against delivering electromagnetic impedance. For a comparative power and voltage levels, it as a rule has cut down cost and less sections while requiring less pc-board zone than a disconnected dc-dc converter. For cut down voltages, non-disconnected buck converters can be used. There are five essential sorts of converter in this non-restricting social affair. They are according to the accompanying.

- Buck Converter
- Boost Converter
- Buck-Boost Converter
- Cuk Converter

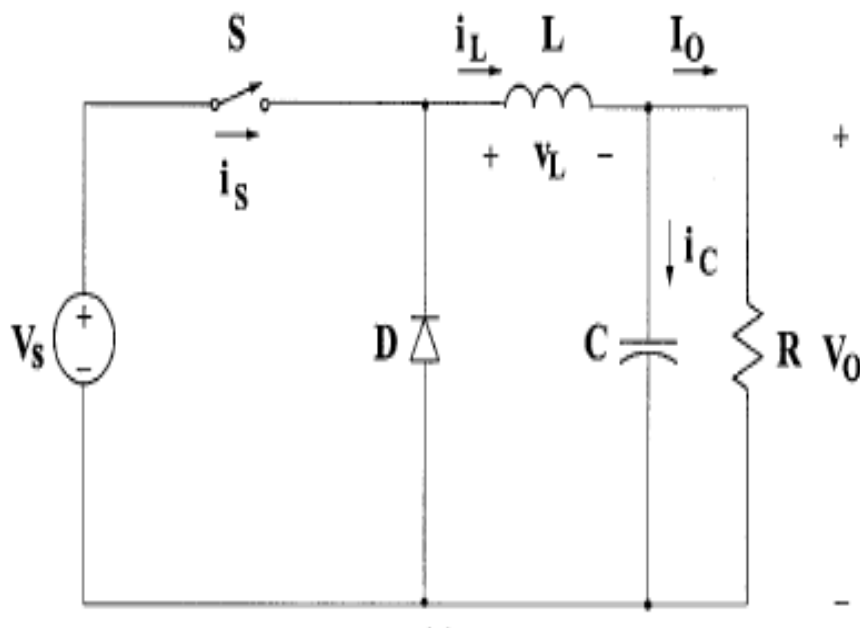
The Buck converter is used for voltage advance down decline, while the Boost converter is used for voltage advance up. The Buck-Boost and Cuk converters can be used for either advance down or advance up, anyway are fundamentally voltage furthest point reversers or 'inverters'. The Charge-pump converter is used for either voltage adventure up or voltage inversion, anyway just in for the most part low power applications.

Buck Converter:

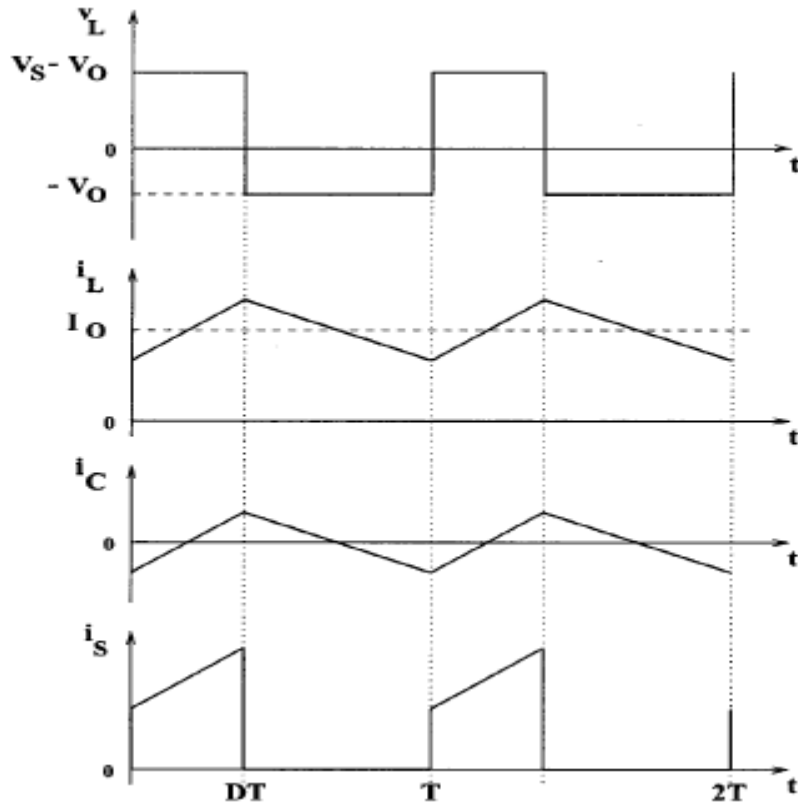
The buck converter is the most comprehensively used dc-dc converter topology in control Management and Microprocessor Voltage Regulator applications. Those applications require snappy load and line transient responses and high profitability over a wide load current range. They can change over a voltage source into a lower controlled voltage. For example, inside a PC system, voltage ought to be wandered down and a lower voltage ought to be kept up. Hence the Buck Converter can be used. In addition, buck converters give longer battery

life to flexible structures that contribute most of their vitality in "stay by". Buck controllers are as often as possible used as switch-mode control supplies for base band mechanized focus and the RF Power Amplifier.

The movement down dc-dc converter, consistently known as a buck converter, is showed up in Fig.3.2(a) It involves dc input voltage source V_S , controlled switch S , diode D , channel inductor L , channel capacitor C , and load restriction R . Normal waveforms in the converter are showed up in Fig.3.2(b) under the doubt that the inductor current is always positive. The state of the converter in which the inductor current is never zero for any time allotment is known as the reliable conduction mode (CCM). It has a tendency to be seen from the circuit that when the switch S is told to the on express, the diode D is transform uneven. Right when the kill S is, the diode practices to help a consistent current in the inductor. The relationship among the data voltage, yield voltage, and the switch commitment extent D can be inf



(a)



(b)

Buck converter: (a) circuit diagram; (b) waveforms.

For the buck converter

$$(V_S - V_O)DT = -V_O(1-D)T \quad (3.1)$$

Hence, the dc voltage transfer function, defined as the ratio of the output voltage to the input voltage is

$$M_V = \frac{V_O}{V_S} = D \quad (3.2)$$

It may be seen from Eq. (3.1) that the yield voltage is constantly littler than the data voltage. The dc-dc converters can work in two evident modes regarding the inductor current i_L . Figure 3.2(b) portrays the CCM in which the inductor current is persistently more perceptible than zero. Right when the commonplace estimation of the yield current is low (high R) and/or the exchanging rehash f is low, the converter may enter the capricious conduction mode (DCM). In the DCM, the inductor current is zero amidst a bit of the exchanging time length. The CCM is favoured for high ability and wonderful usage of semiconductor switches and sit out of gear parts. The DCM might be utilized in applications with exceptional control necessities in light of the way that the dynamic interest of the converter is reduced (the noteworthiness set away in the inductor is zero toward the start and toward the entire of each

exchanging period). It is thrilling to blend these two working modes due to various control figuring's. For the buck converter, the estimation of the channel inductance that picks the cut-off among CCM and DCM is given by

$$L_b = \frac{(1-D)R}{2f} \quad (3.3)$$

For typical estimations of $D = 0.5$, $R=10\omega$, and $f=100$ kHz, the farthest point is $L_b = 25\text{mH}$. For $L > L_b$ the converter works in the CCM. The channel inductor current i_L in the CCM contains a dc part I_O with a superimposed triangular cooling section. The majority of this climate control system section travels through the channel capacitor as a present i_c . Current i_c causes a little voltage swell over the dc yield voltage V_O . To bind the best to-top estimation of the swell voltage underneath a particular regard V_r , the channel capacitance C must be more noticeable than

$$C_{\min} = \frac{(1-D)V_O}{(8V_r L f^2)} \quad (3.4)$$

At $D = 0.5$, $V_r/V_O = 1\%$, $L = 25\text{mH}$, and $f = 100$ kHz, the minimum capacitance is $C_{\min} = 25$ mF.

Conditions (3.3) and (3.4) are the key outline conditions for the buck converter. The info and yield dc voltages (henceforth, the obligation proportion D), and the scope of load protections R are normally controlled by fundamental details. The originator needs to decide estimations of inactive segments L and C , and of the exchanging recurrence f . The estimation of the channel inductor L is ascertained from the CCM=DCM condition utilizing Eq. (3.3). The estimation of the channel capacitor C is gotten from the voltage swell condition Eq. (3.4). For the conservativeness and low conduction misfortunes of a converter, it is alluring to utilize little aloof segments. Conditions (3.3) and (3.4) demonstrate that it very well may be proficient by utilizing a high exchanging recurrence f . The exchanging recurrence is constrained, in any case, by the sort of semiconductor switches utilized and by exchanging misfortunes. It ought to likewise be noticed that estimations of L and C might be modified by the impacts of parasitic segments in the converter, particularly by the equal arrangement obstruction of the capacitor.

Boost Converter:

A Boost Converter i.e. Advance up Converter is a DC-to-DC control converter with a yield voltage more noteworthy than its info voltage. It is a class of exchanged mode control supply containing no less than two semiconductor switches i.e., a diode and a

transistor and something like one vitality stockpiling component i.e., a capacitor, inductor or the two in blend. Channels made of capacitors some of the time in blend with inductors are regularly added to the yield of the converter to decrease yield voltage swell.

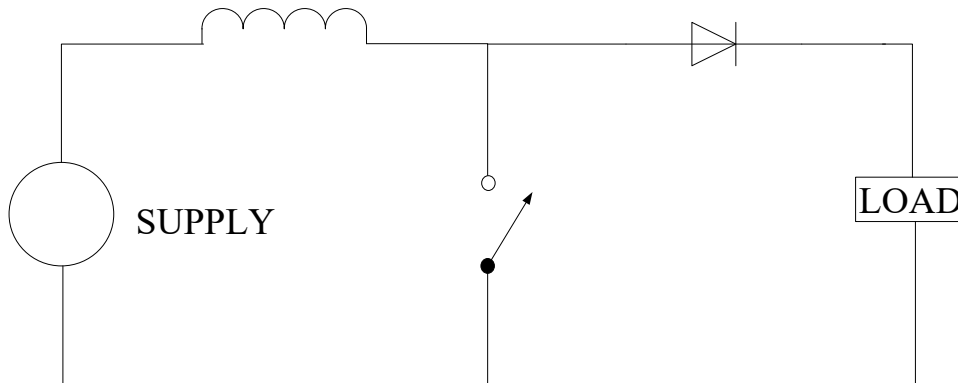


Figure 3.3: Basic schematic of a Boost Converter

Overview of Boost Converter:

Power for the Boost Converter can start from any sensible DC sources, for instance, batteries, daylight based sheets, rectifiers and DC generators. A method that movements one DC voltage to an other DC voltage is called DC to DC change. A lift converter is a DC-to-DC converter with a yield voltage more imperative than the source voltage. A lift converter is now and again rang a phase converter since it "adventures up" the source voltage. Since control $P=VI$ must be talked, the yield current is lower than the source current.

History:

For high adequacy, the SMPS switch must turn on and off quickly and have low mishaps. The happening to a business semiconductor switch in the 1950s addressed a significant perspective that made SMPSs, for instance, the lift converter possible. The huge DC to DC converters were delivered in the mid 1960s when semiconductor changes had ended up being available. The air degree industry's prerequisite for little, lightweight and capable power converters provoked the converter's fast progression.

Traded structures, for instance, SMPS are a test to design since its model depends upon whether a switch is opened or closed. R. D. Focus Brook from Caltech in 1977 dispersed the models for DC to DC converters used today. Focus Brook touched base at the midpoint of the circuit setups for each switch state in a framework called state-space averaging. This change reduced

two systems into one. The new model provoked clever diagram conditions which helped SMPS advancement.

Applications:

Battery energized structures consistently stack cells in course of action to achieve higher voltage. In any case, satisfactory stacking of cells isn't possible in various high voltage applications as a result of nonappearance of room. Lift converters can grow the voltage and diminish the amount of cells. Two battery controlled applications that usage bolster converters are Hybrid Electric Vehicles and lighting structures. A lift converter is used as the voltage increase framework in the circuit known as the 'Joule punk'. This circuit topology is used with low power battery applications and is away for the limit of a lift converter to 'take' whatever is left of the imperativeness in a battery. This essentialness would somehow be misused since the low voltage of a relatively depleted battery makes it unusable for a standard load. This imperativeness would somehow remain unfamiliar in light of the way that various applications don't adequately empower current to course through a pile when voltage reduces. This voltage lessen occurs as batteries end up depleted and is a typical for the inescapable stomach settling agent battery. Since $P=V^2/R$ as well and R tends to be enduring, control open to the pile goes down inside and out as voltage reduces.

Operating standard:

The key decide that drives the Boost Converter are the penchant of an inductor to restrict changes in current. In a lift converter, the yield voltage is continually higher than the information voltage. A schematic of a lift control orchestrate is showed up in Figure 4.2. Right when the switch is closed, current travels through the inductor, which stores imperativeness from the current in an alluring field. In the midst of this time, the switch showings like a short out in parallel with the diode and the stack, so no present streams to the right hand side of the circuit. Right when the switch is opened, the short out is emptied and the store is back in play in the circuit. This addresses a sudden addition in the impedance of the circuit, which, by Ohm's law will ask for either a diminishing in current or a development in voltage. The inductor will have a tendency to contradict such a sudden change in the present, which it does by going about as a voltage source in game plan with the data source, thusly growing the total voltage seen by the right hand side of the circuit and in this way ensuring the present level that was seen when the switch was closed. This is done using the essentialness set away by the inductor.

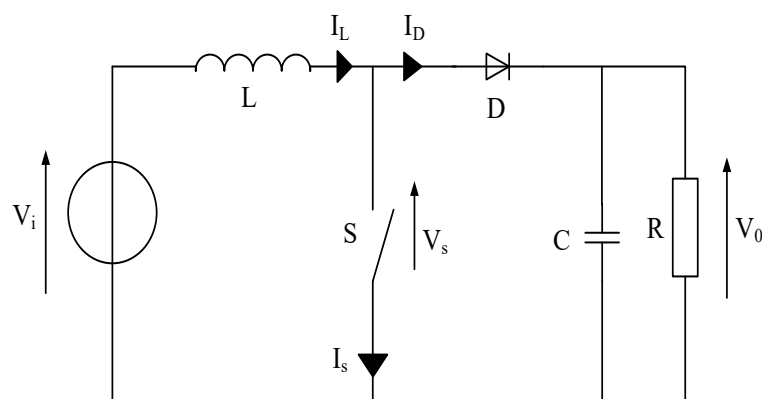
After some time, the imperativeness set away in the inductor will discharge into the right hand side of the circuit, bringing the net voltage pull back.

If the switch is cycled speedy enough, the inductor won't discharge totally trying to charging stages and the pile will constantly watch a voltage more unmistakable than that of the data source alone when the switch is opened. Moreover while the switch is opened, the capacitor in parallel with the pile is charged to this joined voltage. Right when the switch is then closed and the right hand side is shorted out from the left hand side, the capacitor is along these lines prepared to give the voltage and essentialness to the store. In the midst of this time, the blocking diode shields the capacitor from discharging through the switch. The switch ought to clearly be opened again fast enough to shield the capacitor from discharging too much.

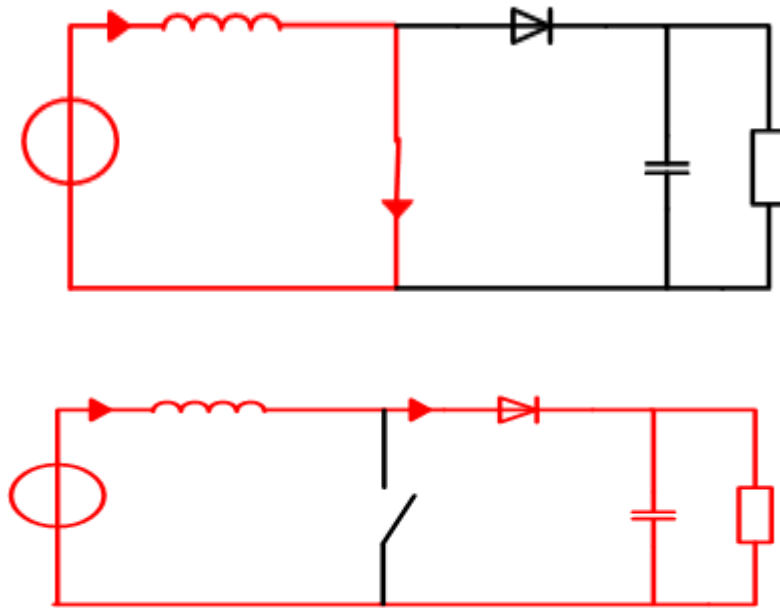
The fundamental lead of a Boost Converter involves two specific states:

- In the On-express, the switch S is closed, realizing a development in the inductor current.
- In the Off-express, the switch S is open and the principle route offered to inductor current is through the fly back diode D, the capacitor C and the store R. This realizes trading the imperativeness amassed in the midst of the On-state into the capacitor.

The data current is the same as the inductor present as can be found in figure 3.4. So it isn't broken as in the buck converter and the essentials on the data channel are released up appeared differently in relation to a buck converter.



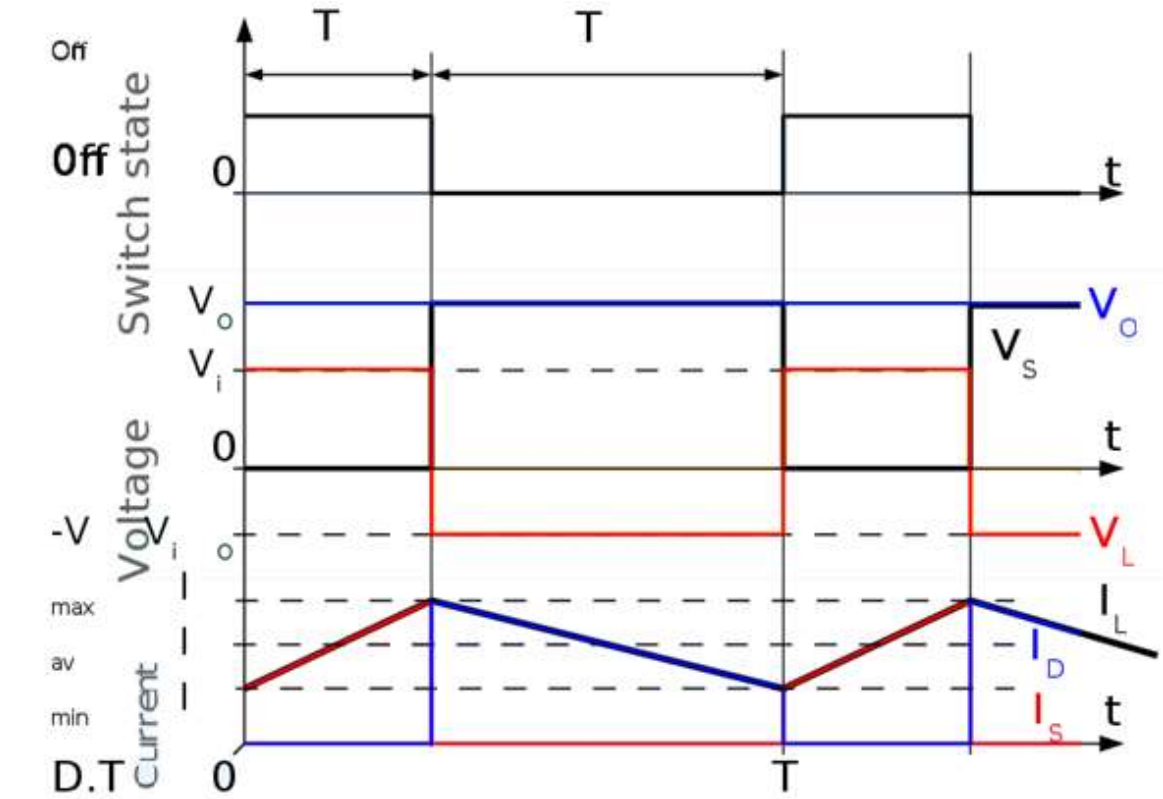
Overall diagram of Boost Converter



The two configurations of boost converter, depending on the state of the switch S

Continuous mode:

Exactly when a lift converter works in consistent mode, the current through the inductor I_L never tumbles to zero. Figure 3.6 exhibits the average waveforms of streams and voltages in a converter working in this mode. The yield voltage can be figured as takes after, by virtue of an immaculate converter i.e. using parts with an impeccable direct working in predictable conditions.



Waveforms of current and voltage in a boost converter operating in continuous mode

Amid the On-express, the switch S is shut, which makes the information voltage V_i show up over the inductor, which causes an adjustment in current I_L moving through the inductor amid a day and age t by the recipe:

D is the obligation cycle. It speaks to the division of the compensation time frame T amid which the switch is ON. Consequently D extends between 0 i.e., Switch is never on and 1 i.e., S is dependably on.

Amid the off-express, the switch S is open, so the inductor current courses through the heap. On the off chance that we consider zero voltage drops in the diode and capacitor sufficiently huge for its voltage to stay consistent, the development of I_L is:

$$V_i - V_o = L \frac{dI_L}{dt} \quad (3.7)$$

Therefore, the variation of I_L during the Off-period is:

$$\Delta I_{Loff} = \int_{DT}^T \frac{(V_i - V_o)dt}{L} = \frac{(V_i - V_o)(1-D)T}{L} \quad (3.8)$$

As we consider that the converter operates in steady state conditions, the amount of energy stored in each of its components has to be the same at the beginning and at the end of a commutation cycle. In particular, the energy stored in the inductor. The above expression shows that the output voltage is always higher than the input voltage as the duty cycle goes from 0 to 1 and that it increases with D , theoretically to infinity as D approaches 1. So, this converter is sometimes referred to as a step-up converter.

Discontinuous mode:

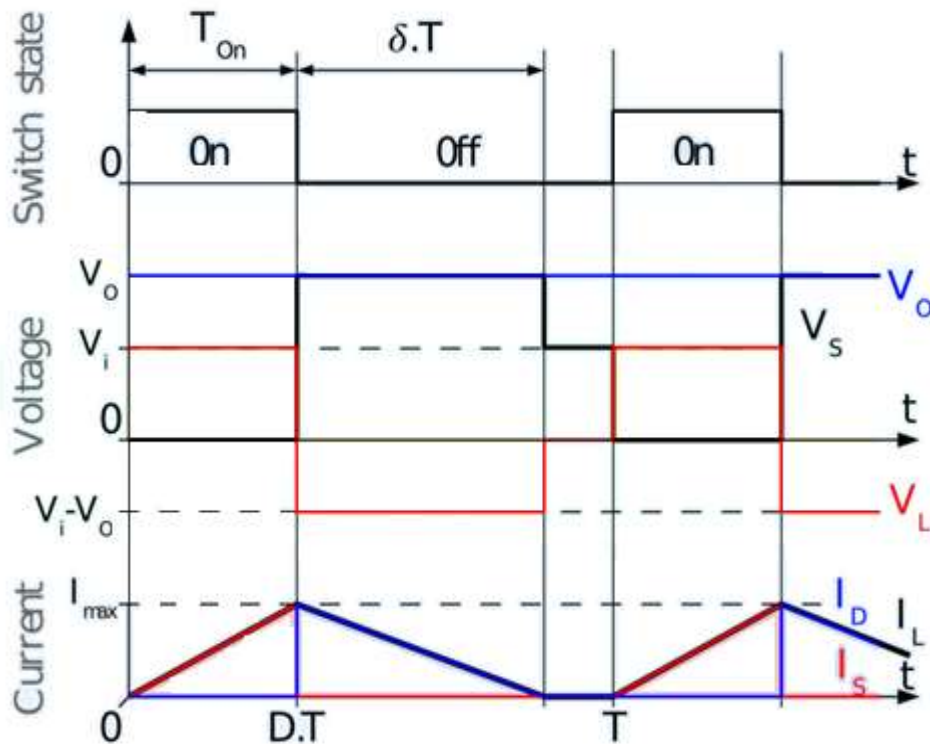


Figure 3.7: Waveforms of current and voltage in a boost converter operating in discontinuous mode.

On the off chance that the swell abundance of the current is too high, the inductor might be totally released before the finish of an entire replacement cycle. This regularly happens under light loads. For this situation, the current through the inductor tumbles to zero amid part of the period as appeared in figure 3.7. Albeit slight, the distinction strongly affects the yield voltage condition.

It very well may be ascertained as takes after:

As the inductor current toward the start of the cycle is zero, its most extreme esteem I_{LMax} at $t=DT$ is given by

$$I_{LMax} = \frac{V_i DT}{L} \quad (3.14)$$

During the off-period, I_L falls to zero after δT :

$$I_{LMax} + \frac{(V_i - V_o)\delta T}{L} = 0 \quad (3.15)$$

Using the two previous equations, δ is:

$$\delta = \frac{V_i D}{V_o - V_i} \quad (3.16)$$

The load current I_o is equal to the average diode current I_D . From figure 3.4, the diode current is equal to the inductor current during the off-state. Therefore the output current can be written as:

$$I_o = \bar{I}_D = \frac{I_{LMax}}{2} \delta \quad (3.17)$$

Replacing I_{Lmax} and δ by their respective expressions yields:

$$I_o = \frac{V_i DT}{2L} \cdot \frac{V_i D}{V_o - V_i} = \frac{V_i^2 D^2 T}{2L(V_o - V_i)} \quad (3.18)$$

Therefore, the output voltage gain can be written as follows:

$$\frac{V_o}{V_i} = 1 + \frac{V_i D^2 T}{2LI_o} \quad (3.19)$$

Appeared differently in relation to the statement of the yield voltage for the relentless mode, this explanation is significantly more convoluted. Furthermore, in broken movement, the yield voltage gain depends upon the commitment cycle and in addition on the inductor regard, the data voltage, the trading repeat and the yield current.

Buck-Boost Converter:

A non-withdrew (transformer-less) topology of the buck-bolster converter is showed up in Fig.3.8. The converter includes dc input voltage source V_S , controlled switch S , inductor L , diode D , channel capacitor C , and load check R . With the switch on, the inductor current additions while the diode is kept up off. Exactly when the turn is executed, the diode gives a path to the inductor current. Note the furthest point of the diode that results in its current being drawn from the yield.

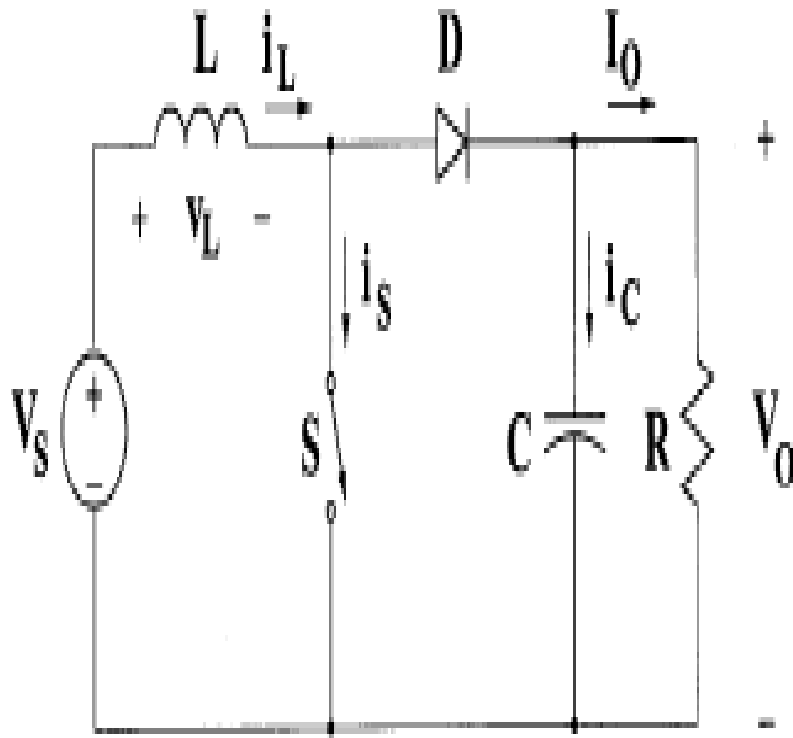
The buck-boost converter waveforms are depicted in Fig.3.9. The condition of a zero volt-second thing for the inductor in reliable state yields

$$V_s DT = -V_o(1-D)T$$

Hence, the dc voltage transfer function of the buck-boost converter is

$$M_v = \frac{V_o}{V_s} = -\frac{D}{1-D}$$

The output voltage V_o is negative with respect to the ground. Its magnitude can be either greater or smaller (equal at $D = 0.5$) than the input voltage as the name of the converter implies.

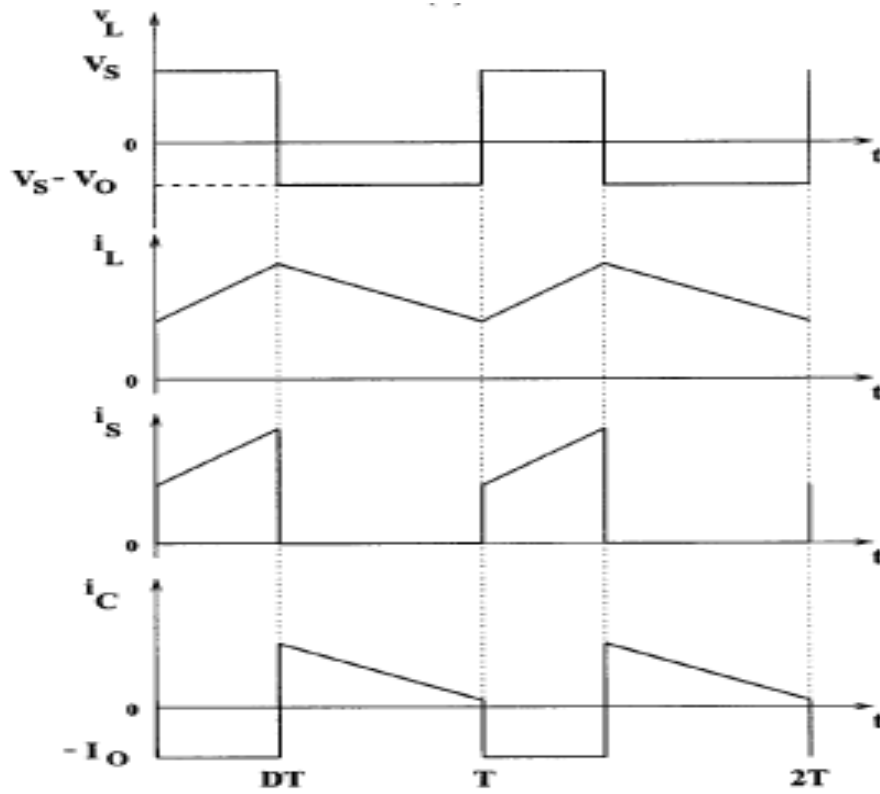


buck-boost converter

The value of the inductor that determines the boundary between the CCM and DCM is

$$L_b = \frac{(1-D)^2 R}{2f}$$

The structure of the output part of the converter is similar to that of the boost converter (reversed polarities are the only difference). Thus, the value of the filter capacitor can be obtained from Eq



buck-boost converter waveform

Chapter-4

UPQC MODULE

Although the main role of UPQC systems is performing series–parallel compensation, so that they can simultaneously act as series APF, compensating for mains voltages, as well as acting as P-APF, compensating the load currents, in [21] experimental results of the single-stage PV system integrated with UPQC performs only the function of a dynamic voltage restorer. In this case, only the disturbances of the grid voltages are compensated. In [22], a double-stage PV system integrated with the UPQC, named SPV-UPQC-P, has been evaluated only through computer simulations. However, this system only compensates reactive power of the load and unbalances of the grid voltage. Thus, the suppression of grid voltage as well as load current harmonics has not been taken into account. Another application in which the PV system is integrated with the UPQC is presented in [23]. In this application, the system can operate as a grid forming in an acmicrogrid [19], since different types of DG sources (PV, wind, and others), as well as energy storage systems, can be used as grid-forming units in an islanded microgrid [29]. However, transients/disturbances could be observed in the voltages that fed the load when the system was transferred from the grid-connected operation mode to the grid-islanded operation mode. This happens because the parallel converter of the UPQC needs to change its control mode from current source to voltage source. The same effect occurs when the system returns to operate in the grid-connected mode, because the parallel converter must be controlled again as the current source.

In [24], the system called the PV-UPQC has been presented. It can operate in both three-phase three-wire and three-phase fourwire EPS. Since the parallel converter is voltage controlled, so that balanced and regulated voltages can be provided to the load, there is no need to change its control mode when the system operates as grid forming in an ac microgrid. In other words, the parallel converter is voltage controlled in both grid-connected and grid-islanded modes. On the other hand, the mentioned system can also operate either as grid feeding or grid supporting [19] in an ac microgrid, since the control mode of the parallel converter can also be switched to operate from voltage source to current source.

On the other hand, in [24], studies related to stability analysis, detailed study related to active and apparent power flows and mainly the sizing and protection of the power converters that

compose the PV-UPQC system, have not been addressed. Thereby, additional research advances and contributions are presented in this paper, as follows.

- 1) A complete study involving the power flow through the PV-UPQC system for obtaining the overall understanding of the system working under several operation modes is performed. This study represents an important and useful methodological tool for designing the power converters properly. It is supported by means of an extensive number of sizing curves and allows the designer an effective power converters sizing.
- 2) A strategy to avoid over power rating of the series and parallel power converters is implemented. This strategy is needed in order to establish the priority of the power flow through the converters, since the PV-UPQC system performs, simultaneously, grid active power injection (energy produced from the PV system), as well as the power-line conditioning.
- 3) The stability analysis of the PV system is performed. In the context of a UPQC, the study involving the ability of the series and parallel converters to remain stable even in the occurrence of disturbances in both the load currents and grid voltages has never been addressed before in the literature and appears as an important and necessary subject to be discussed. Furthermore, it is checked if the system stability is affected or not by different grid impedance characteristics.
- 4) The PV-UPQC system is also tested in grid-islanded operation. This operation mode allows exploring new aspects related to the multifunctionality of the PV-UPQC system.

This paper is organized as follows. Section II contains the description of the PV-UPQC system, as well as the strategies used to control the series and parallel inverters in grid-connected and grid-islanded operation modes. Section III contains detailed descriptions of the active and apparent power flows through the PV-UPQC resulting normalized curves for a proper sizing of the series and parallel neutral point clamped (NPC) inverters. In addition, power limitation strategies applied to the power converters are presented. Section IV contains the descriptions of system stability analysis, while in Section V, the strategy developed to avoid the over power rating of the parallel NPC inverter is tested. In Section VI, both static and dynamic performances of the system are evaluated according to the experimental results. Finally, Section VII presents the conclusions.

DESCRIPTION OF THE PV-UPQC SYSTEM

Fig. 1 shows the power circuit of the three-phase single-stage PV-UPQC system. The series and parallel converters consist of three-level NPC inverters. The series inverter is connected to the grid by means of three coupling transformers connected in series with L filters, represented by the inductances $L_{sc\ abc}$ in series with their respective internal resistances $R_{sc\ abc}$. The parallel inverter is connected to the point of common coupling 2 (PCC2) through LC filters represented by the inductances $L_{pc\ abc}$ in series with their respective internal resistances $R_{pc\ abc}$, in addition the capacitances $C_{p\ abc}$. The dc bus is composed of the capacitors C_{dc1} and C_{dc2} connected in parallel with the PV array consisting of a string with 20 series-connected PV panels.

The operating voltage of the PV system is determined by the MPPT algorithm called perturb and observe (P&O) [16]. Thus, the maximum amplitude of the dc-bus voltage ($v^*_{dc\ max}$) is around 600 V, allowing the PV system to operate in MPP at standard test conditions (STC). On the other hand, the minimum

The block diagram of the proposed system is illustrated in Fig. 1. A three-phase, three-wire grid is integrated with the PV, WE, ESS, and EVs through a phase inverter. The line-to-line voltage of the system and frequency is 254.4 V and 50 Hz, respectively. The voltage disturbance is observed due to non-linear loading, voltage swell, and voltage sag. To overcome this issue, a UPQC is designed and connected to the grid for voltage regulation. Two WE systems having 30 kW power output each are connected which totals the generated power through WE is 60 kW. The ESS system contains a battery bank connected to the grid having 36 series connected 12V, 100 Ah batteries. The capacity of the ESS system is 40 kWh and is connected to the grid through a bidirectional converter. The capacity of the PV system is 100 kW. The PV and WE systems are connected to the grid through boost converters. In this work, EVs are also considered each having a capacity of 29 kW. The FLC and ANN-based MPPT algorithms are designed for both PV and WE systems. The problem statement, contributions and objectives of this research work are highlighted in the next section.

From the literature survey, it can be inferred that the power quality improvement of PV, WE, PV-ESS, and WE-ESS systems have been carried out using various techniques such as UPQC, UPQC-PI, UPQC-FLC, and UPQC-NN. Nowadays, EVs are also becoming a part of the PV-ESS system and need to be considered in power quality improvement issues along

with the PV-ESS system. The function of EVA is to manage the power flow between EV, ESS, and PV systems. Thus it can also act as an essential component for improving the quality of power. Therefore, the main contribution of this work focuses on the power quality improvement of the PV-WE-ESS-EV system. The roles of EVs and UPQC on power enhancement and grid stabilization are highlighted. It is observed that during the peak load demand in the grid, the excess power demand is managed from the ESS and EVs. The condition of the worst case is also considered that is when the availability of power from RESs is less and the grid power does not suffice to satisfy the load demand. During such a situation, the EVA manages the power demand of the grid-connected load by arranging EVs. The EVA controller operates and manages power from EVs to grid or grid to EVs with the help of a load forecasting system, power generation forecasting system of RESs, and nearby charging stations. The use of UPQC along with EVs to the grid system provides improved power and voltage stabilization which is shown in detail in this work. It means that the main objective of implementing EVA is to manage the utilization of EVs as a temporary battery storage system which is transportable and thus, the stored energy can be also utilized for transportation purpose when it is not required to be used as ESS. The FLC technique is used in this work to control the proposed technique and the results obtained are compared with the ANN approach. The FLC is refreshed in every 0.01 second for the grid stabilization and power improvement.

Power enhancement of PV-WE-ESS-EV system.

- _ Reduction of THD of grid voltages and currents.
- _ To obtain good voltage regulation at various loading conditions.
- _ To avail enough power to fulfil the load demand under the worst situation in which RES is not available for power generation.
- _ To utilize the EVs as ESS during excess availability of power and as power supplier during peak load demand of the grid.

The MPPT algorithm plays an important contribution in generating maximum power at various weather conditions.

In this research work, the fuzzy and ANN-based controllers are developed in a MATLAB environment for PV and WE systems. Finally, the simulation results of both the controllers are compared to select the best one. The EV batteries designed are of 29 kW capacity each (12 V). The battery management system and load balancing with EVs are accomplished by using EVA. The EVA is developed through Fuzzy and ANN for its performance analysis. The grid integration of the hybrid PV-WE system is still a challenging point due to its non-linear characteristics of the power generated through the RESs. The design of controllers for grid integration contains a phase lock loop, voltage regulator, and current regulator, as proposed in this work. The voltage source converter based controller plays a major part in synchronizing RESs into the grid. The proposed control technique includes PV, WE, ESS, and EVs, which are connected to the grid along with the UPQC-FLC control of the PV-WE-ESS-EV system. This technique cumulatively improves the power quality and reduces the THD values of voltage and current of the grid below the threshold limit provided by ANSI/IEEE-519 (1992) standard. The FLC and ANN techniques are implemented in the PV-WE-ESS-EV system and the outcomes are compared to get the best technique for achieving improved quality of power, grid voltage, and frequency stabilization. The techniques available in the literature and system of application are summarized.

CHAPTER-5

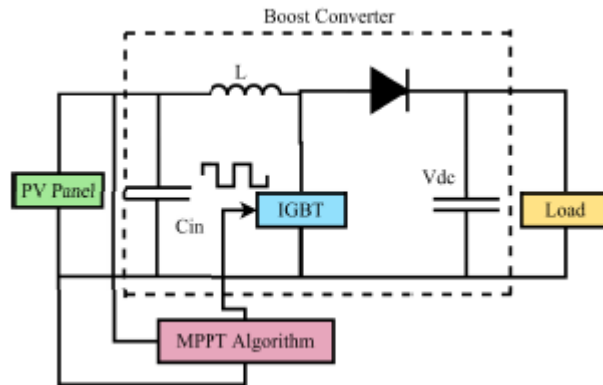
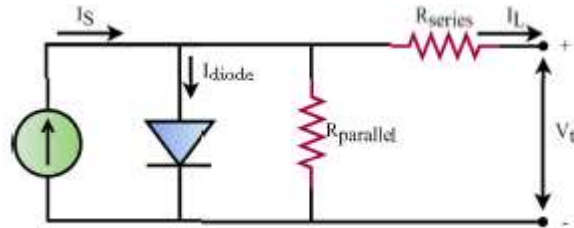
PROPOSED SYSTEM

The work focuses on the power enhancement and grid stabilization of RESs with ESS-EV and non-linear loads. In this Section, the mathematical modeling of solar PV and WE systems are discussed. The MPPT algorithms and design procedure are also provided in this Section.

MODELLING OF SOLAR PV SYSTEM

In the literature, many researchers discussed the modeling of PV and designing MPPT algorithms to extract maximum possible power from the PV panel [40]_[45]. The circuit diagram of a simple PV system is shown in Fig. 2. The mathematical equation for the I-V characteristics of the PV panel has been expressed in detail by the authors in [46] and is represented in equation 1. In equation 1, I_L is the current output from the panel, I_{diode} is the current flowing through the diode, I_S is the current from PV panel, the thermal voltage is represented by V_T , c is the diode constant, series and parallel branch resistances are represented by R_{series} and $R_{parallel}$, respectively, and I_{sat} is the reverse saturation current.

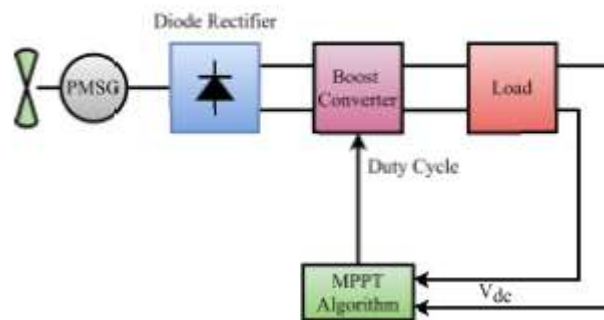
$$I_L = I_S - I_{sat} \left[\exp\left(\frac{V_t + R_{series}I_L}{cV_T} - 1\right) - \frac{V_t + R_{series}I_L}{R_{parallel}} \right]$$



The block diagram of the PV panel with MPPT algorithm and boost converter is shown in Fig. 3. The drawbacks of conventional MPPT algorithm, including poor performance under

non-uniform shading, selection of local maximum point of power under partial irradiation are overcome using the FLC-based MPPT algorithm [47]. In the recent era, FLC is the most preferred technique for the MPPT algorithm. Therefore, FLC-based MPPT is designed in this research study. For controlling a system, process, or a device, fuzzy logic is very useful, simple, robust, and adaptive. It has the capability of making decisions. The block diagram of the fuzzy logic-based MPPT controller is shown in Fig. 4. The first stage in FLC is the fuzzy cation which converts the values and parameters into the fuzzified form using a function of transformation. The next stage is the interfacing in which the fuzzy rules are set. The last stage is the defuzzification in which the fuzzily parameter is again converted into the crisp form. The parameters used in FLC-based MPPT are PV voltage and current. The power and change in power are evaluated using voltage and current parameters. The controller is a type-2 FLC whose membership function and fuzzy set are given in any form from equations 2,3, and 4 as explained in above.

The WE has a permanent magnet synchronous generator (PMSG) for power generation from the wind turbines. The PMSG is used because of its advantages over doubly fed induction generator (DFIG) including greater efficiency, lower cost, easily controllable, and reliable. The mathematical modelling of WE and FLC-based MPPT for the WE system have been provided in many existing literatures



Chapter-6

MATLAB/SIMULATION

Matlab is a remarkable language for specialized handling. It includes computation, assumption, and also shows in a straightforward to-utilize problem where concerns as well as setups are interacted in all-natural mathematical paperwork. Ordinary usages integrate Math and also computation Algorithm enhancement Data protecting Modeling, recreation, and also prototyping Data exam, examination, and also assumption Scientific as well as developing images Application development, consisting of visual UI structure. Matlab is an user-friendly structure whose crucial details part is an exhibition that does not need dimensioning. This allows you to care for many specialized signing up problems, specifically those with network as well as vector meanings, in a percentage of the moment it would certainly require to make up a program in a scalar no smart language, for instance, C or Fortran. The name matlab stays for grid study center. Matlab was at first made up to provide easy accessibility to structure programs produced by the linpack as well as eispack endeavors. Today, matlab electric motors settle the lapack and also blas collections, putting the reducing side in shows for grid estimation. Matlab has actually progressed over a season with payment from countless customers. In university problems, it is the common educational device for beginning and also thrust programs in math, structure, as well as scientific research.

In sector, matlab is the gadget of choice for high-efficiency study, improvement, and also evaluation. Matlab highlights a team of additional application-particular setups called device stock. Essential to the majority of customers of matlab, device stockpile allow you to find out and also use specific advancement. Device areas are extensive build-ups of matlab capabilities (M-records) that extend the matlab problem to care for details courses of problems. Areas where device stockpile come include flag handling, control structures, neural systems, cosy reasoning, wavelets, recreation, as well as various others. The matlab structure consists of 5 key components: Development Environment. This is the plan of tools and also workplaces that help you use matlab capabilities and also records. A significant great deal of these tools are visual UIs. It integrates the matlab workplace and also Command Window, a cost background, a supervisor and also debugger, and also programs for study urge, the work area, documents, and also the search method.

The matlab Mathematical Function Library. This is a remarkable event of computational computations going from primary capabilities, just like accumulation, sine, cosine, as well as intricate mathematics, to even more polished abilities like grid in reverse, structure Eigen confidences, Bessel abilities, and also fast Fourier modifications. The matlab Language. This is an uncommon state lattice/cluster language with control stream expressions, capabilities, details frameworks, input/yield, and also inquiry positioned programs highlights. It allows both "shows in the little" to swiftly make to service throw out jobs, as well as "programs in the significant" to earn large as well as complicated application programs.

Matlab has wide workplaces for revealing vectors and also grids as graphes, as well as on top of that discussing and also publishing these representations. It includes uncommon state abilities for two-dimensional and also three-dimensional details depiction, image handling, motion, as well as intro pictures. It also includes low-level capabilities that allow you to totally customize the existence of pictures and also along with produce coating visual UIs on your matlab applications.

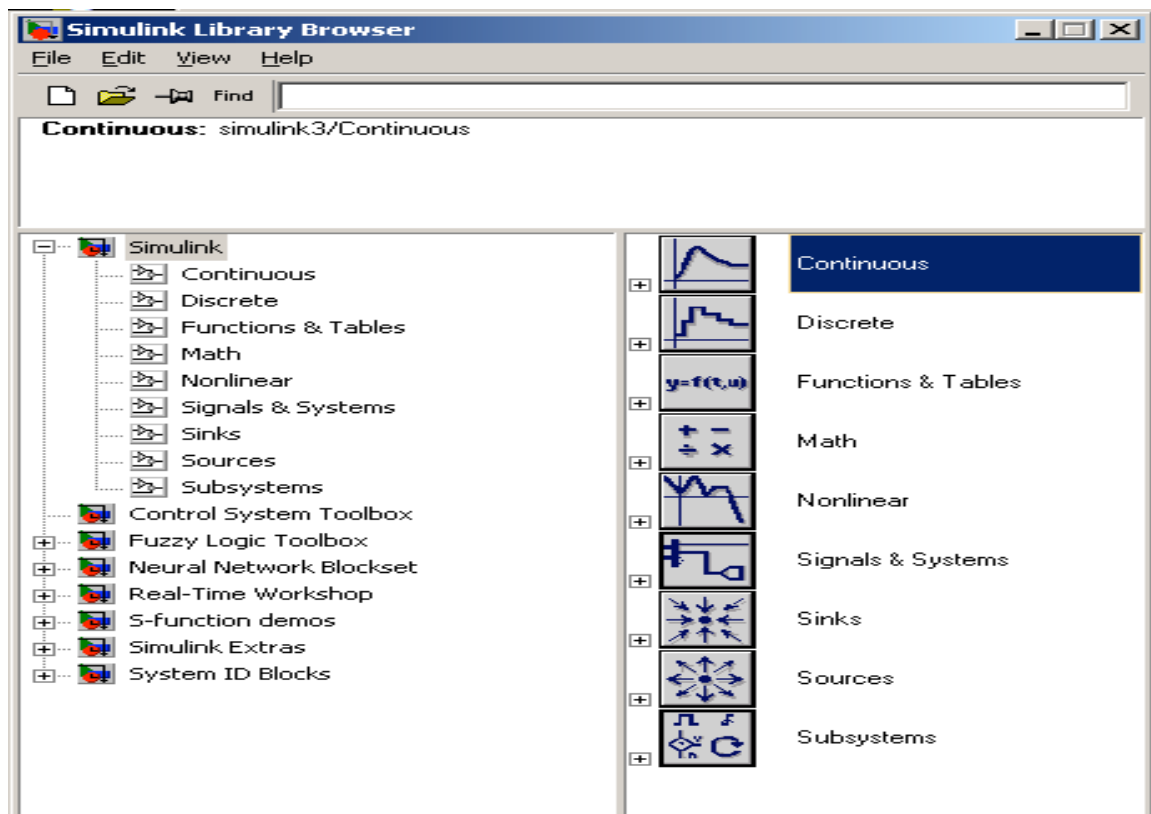
The matlab Application Program Interface (API). This is a collection that allows you to make up C as well as Fortran programs that work together with matlab. It includes workplaces for calling routines from matlab (vibrant linking), calling matlab as a computational electric motor, as well as for reading and also making up MAT-records.

SIMULINK:

Simulink is a product add-on to matlab which is a mathematical tool made by The Math functions,(<http://www.mathworks.com>) an organization prepared in Natick. Matlab is powered by large mathematical assessment restriction. Simulink is a mechanical setting up utilized to seemingly configure a vibrant framework (those managed by Differential problems) as well as consider outcomes. Any type of basis circuit, or control system for a vibrant framework could be functioned by utilizing basic structure squares open in Simulink Libraries. Distinct device stock for various structures, as an example, Fuzzy Logic, Neural Networks, dsp, Statistics and so on are open with Simulink, which revamp the handling pressure of the tool. The policy benefit is the visibility of organizations/building squares, which stay clear of the requirement of producing code for very little mathematical systems.

Concept of banner as well as factor stream:

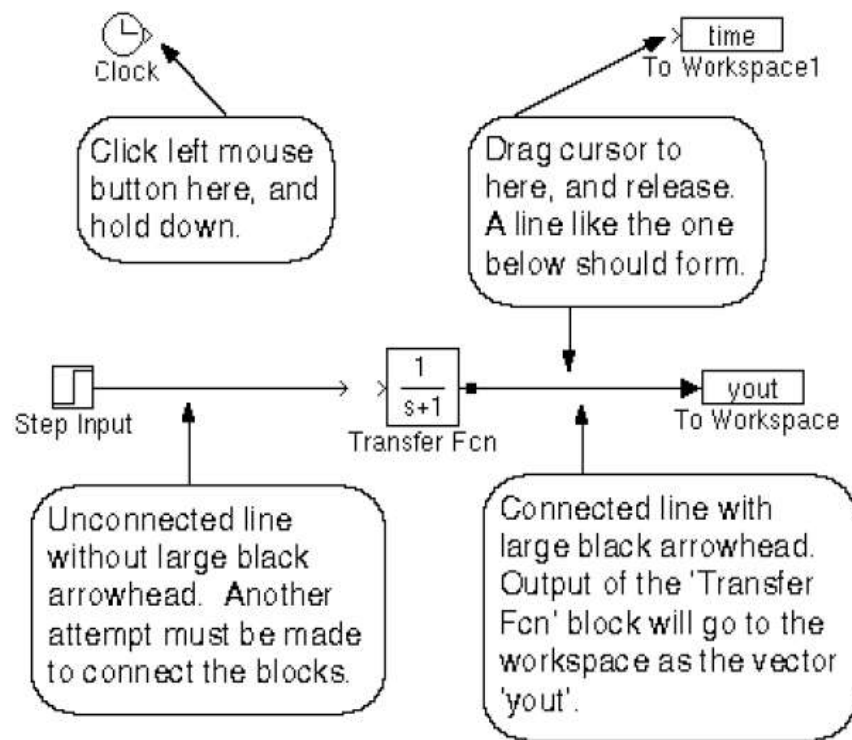
In Simulink, data/information from numerous squares are sent out to an additional square by lines interfacing the appropriate squares. Indications could be supplied and also sustained right into squares dynamic/static). Information could be enhanced right into restrictions. Information would certainly after that have the ability to be unloaded right into sinks, which might be enhancements, shows up or might be conserved to a document. Information could be relevant start with one square after that into the adhering to, could be fanned, multiplexed and so on. In proliferation, information is looked after and also traded specifically at distinct occasions, considering that all PCs are distinct systems. This way, a reenactment time action (generally called a synchronisation time action) is essential, as well as the choice of that activity is regulated by the fastest elements in the duplicated framework.



Simulink library browser

Connecting blocks:

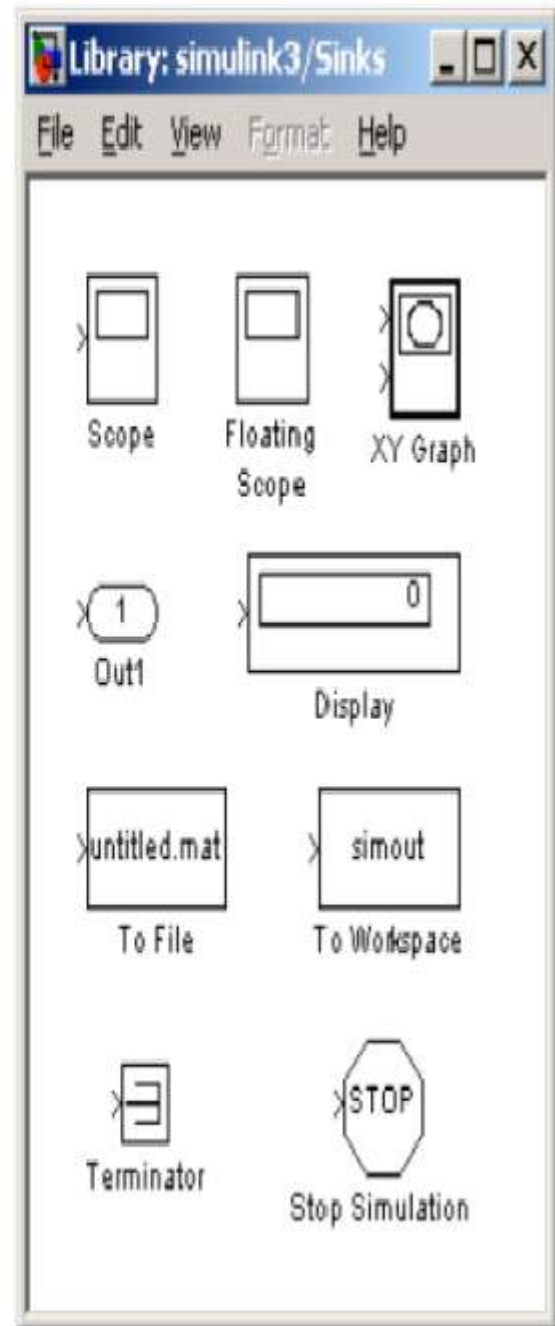
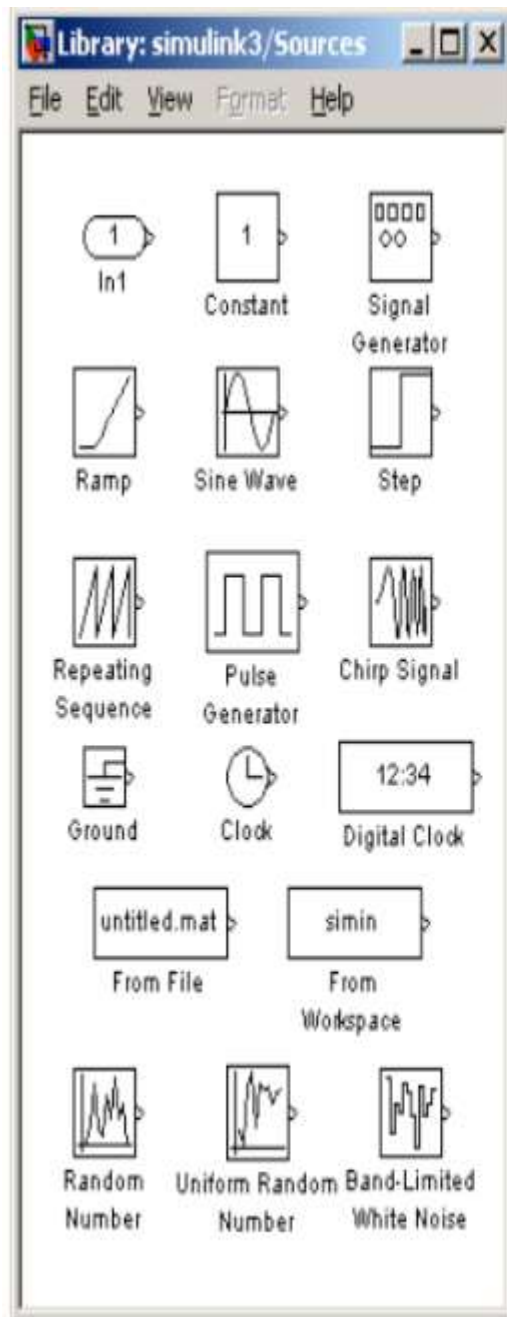
To connect blocks, left-click and drag the mouse from the output of one block to the input of another block.



Connectung blocks

Sources and sinks:

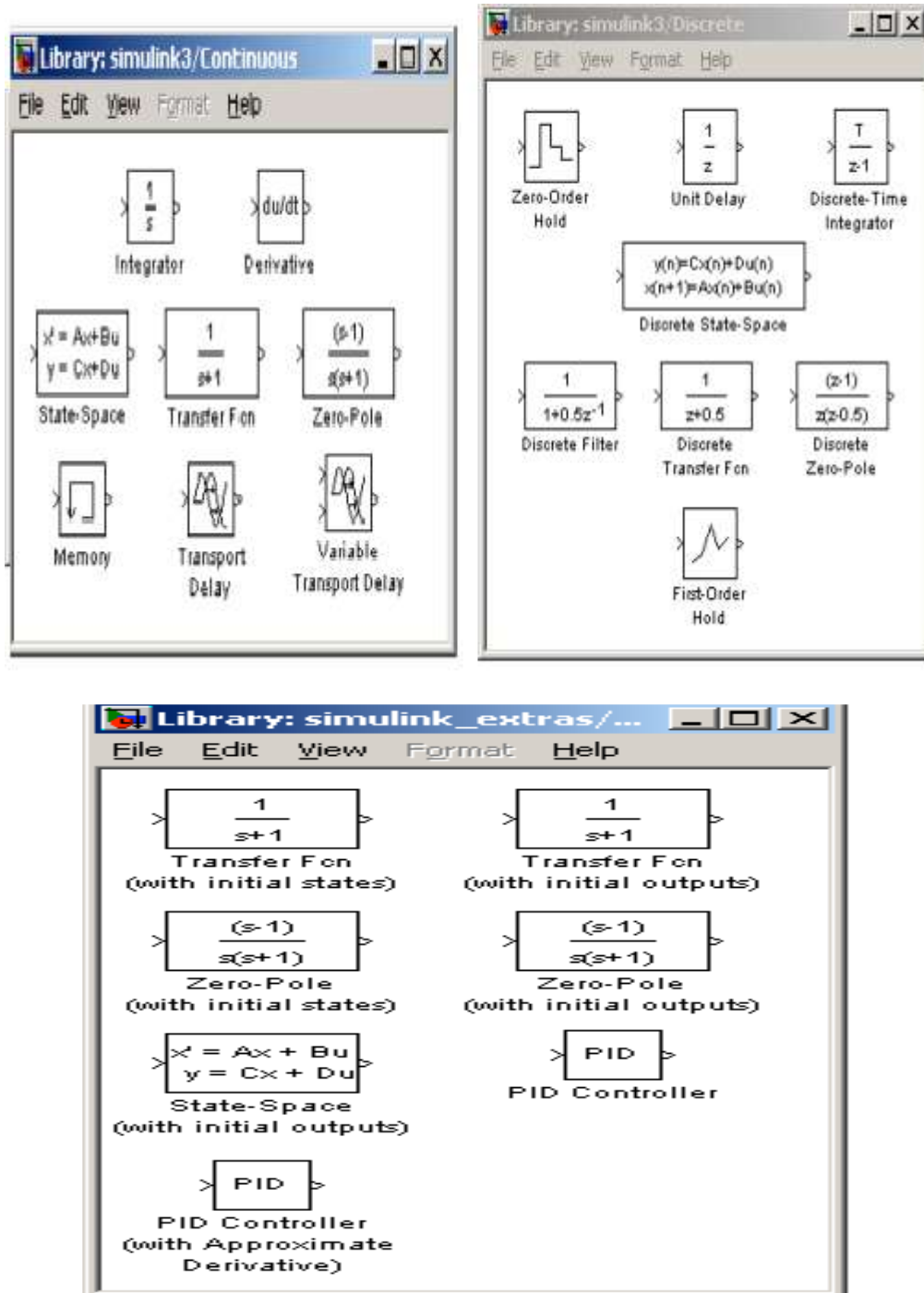
The resources collection consists of the roots of information/flags that would certainly utilize in a vibrant structure recreation. One could should use a regular info, a sinusoidal wave, a phase, a reworking sequence, for instance, a heart beat prepare, an incline etc. One maybe ought to evaluate upsetting impact effects, and also could use the uneven flag generator to re-enact turmoil. The clock may be made use of making a duration document for outlining functions. The ground can be made use of to connect with any type of extra port, to avoid notification messages revealing removed ports. The sinks are squares where signals are finished or at some point used. Generally, we would certainly should save the succeeding info in a document, or a network of variables. The info might be revealed and even do away with to a record. The quit square can be used to quit the recreation if the payment to that square (the flag being sunk) is non-zero. Number 3 shows the easily accessible squares in the resources as well as sinks collections. Extra indications should be finished, to neutralize reproaches concerning removed indicators.



Sources and sinks

Continuous and discrete systems:

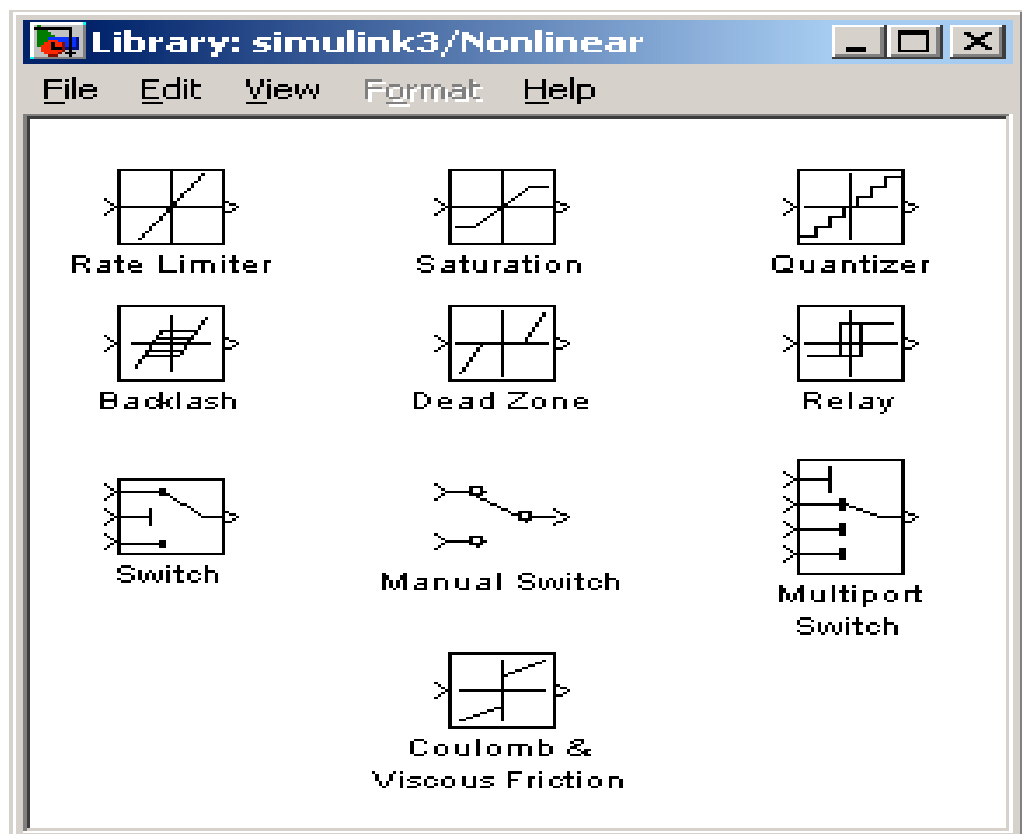
Every powerful framework can be investigated as nonstop or discrete time frameworks. Simulink enables you to speak to these frameworks utilizing exchange capacities, combination squares, defer squares and so on.



continuous and discrete systems

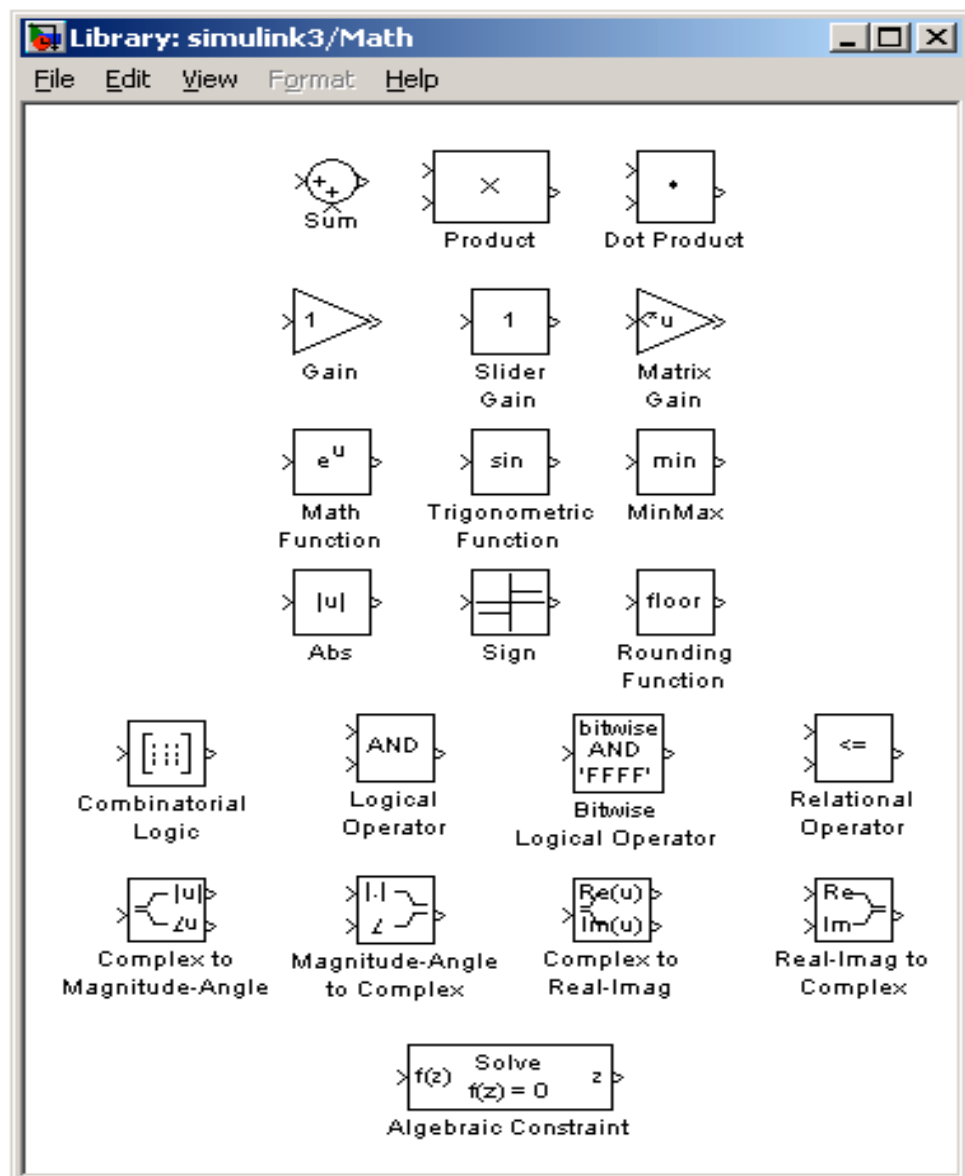
Non-linear operators:

A concept chosen point of view of using devices, for instance, Simulink is the capability to recreate non-straight structures and also land at outcomes without resolving realistically. It is remarkably tough to land at an analysis solution for a structure having non-linear connections, for instance, immersion, details exchange job, as well as limited variety prices and so forth. In Simulation, because structures are analyzed making use of cycles, non-linearity are not a blockage. One such can be an immersion block, to show a physical restraint on a specification, as an example, a voltage flag to an engine and so forth. Hand-operated buttons are practical while trying re-enactments with numerous instances. Buttons are exactly what may also be called if-then pronouncements in programs.

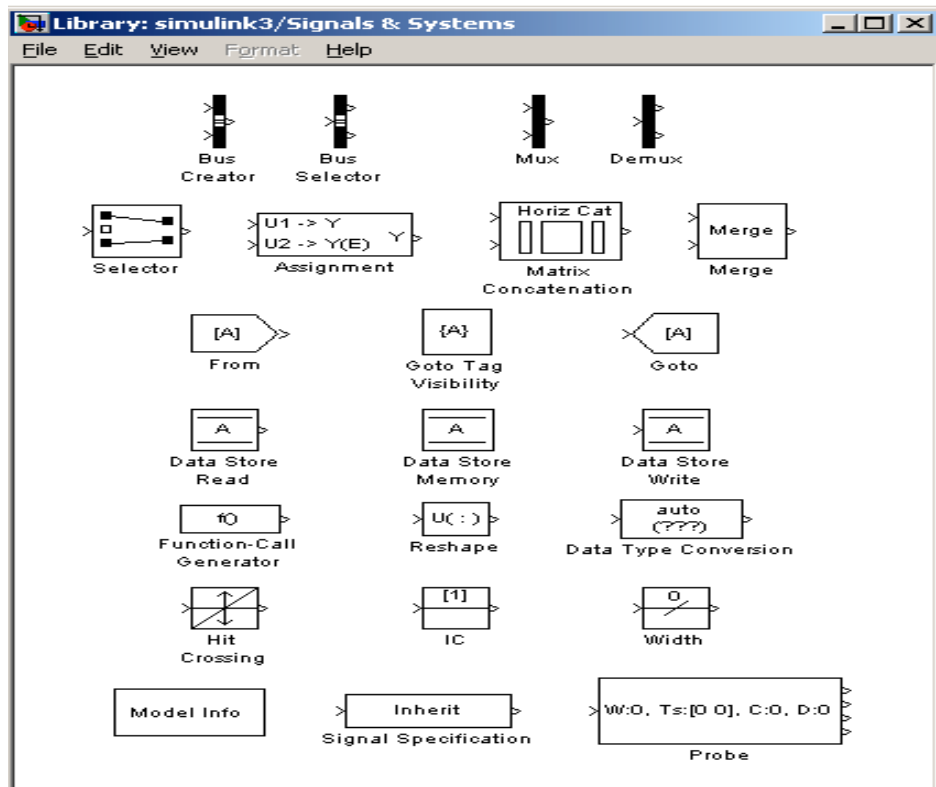


simulink blocks

Mathematical operations:



Simulink math blocks



signals and systems

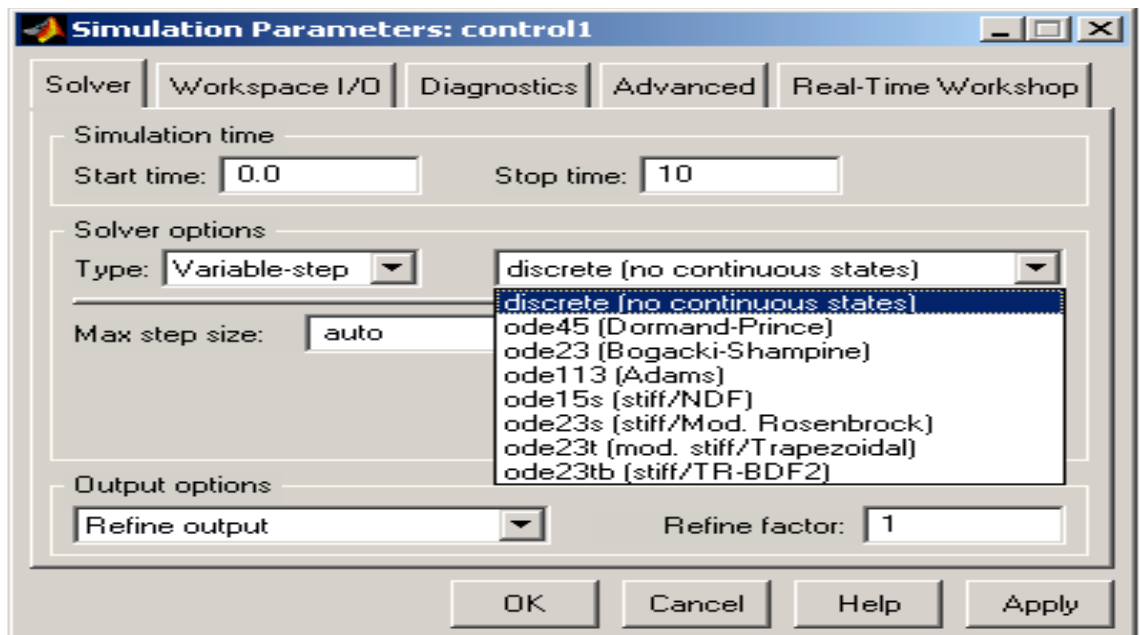
SIGNALS & DATA TRANSFER:

In baffled square graphs, there could arise the should exchange info beginning with one component after that into the following section of the square. They may be in numerous subsystems. That flag might be unloaded right into a goto square, which is used to send out signals beginning with one subsystem after that into the following. Multiplexing urges us leave mess due to severe ports, and also makes structure (section/push) depiction easier.

Making subsystems: Drag a subsystem from the Simulink Library Browser as well as put it in the moms and dad square where you could intend to hide the code. The sort of subsystem trusts the factor for the square. Generally one will certainly use the typical subsystem nonetheless various subsystems could be chosen. As an example, the subsystem could be a triggered square, which is encouraged simply when a trigger flag is obtained. Open (dual faucet) the subsystem as well as make input/yeild PORTS, which relocate signals right into as well as from the subsystem. The info and also return ports are made by carrying them from the Sources and also Sinks magazines independently. At the factor when ports are made in the subsystem, they normally make ports on the external (moms and dad) square. This thinks about connecting the appropriate indications from the moms and dad square to the subsystem.

Setting recreation specifications :

Running a recreation in the COMPUTER reliably calls for a mathematical treatment to comprehend a differential problem. The structure could be resembled as a constant structure or a distinct structure because the squares inside. The recreation start and also quit time could be suggested. In case of variable development dimension, the smallest as well as most significant development dimension could be established. A Fixed breakthrough dimension is recommended and also it takes into account purchasing time to a specific variety of concentrates, by doing this regulating the period of the details vector. Leisure action procedure have to be picked because the components of the structure. A cozy treatment might necessitate a phase dimension of a number of minutes, nonetheless a DC engine in the structure could be extremely fast as well as could need a phase dimension of a few nanoseconds.



simulation parameters

Chapter-7

SIMULATION RESULTS

The proposed work focuses on the power enhancement and harmonic reduction for the PV-WE-ESS-EV system connected to the grid. The UPQC is able to handle the non-linear and different undesired types of loads. The whole system is refreshed and checked by FLC in every 0.01 second. In this Section, the outcomes of the two techniques are discussed. The first is based on FLC, and the second one is based on the ANN approach. It is observed that FLC is giving better results as compared to ANN to achieve maximum power from the PV-WE system, to get good voltage regulation, and to reduce the THD values of voltages and currents. Also, ANN requisites learning data and time to perform better and accurate. The learning of ANN techniques is time-consuming and also need various historical data to learn from them.

A. ROLE OF EVA IN PV-WE-ESS-EV SYSTEM

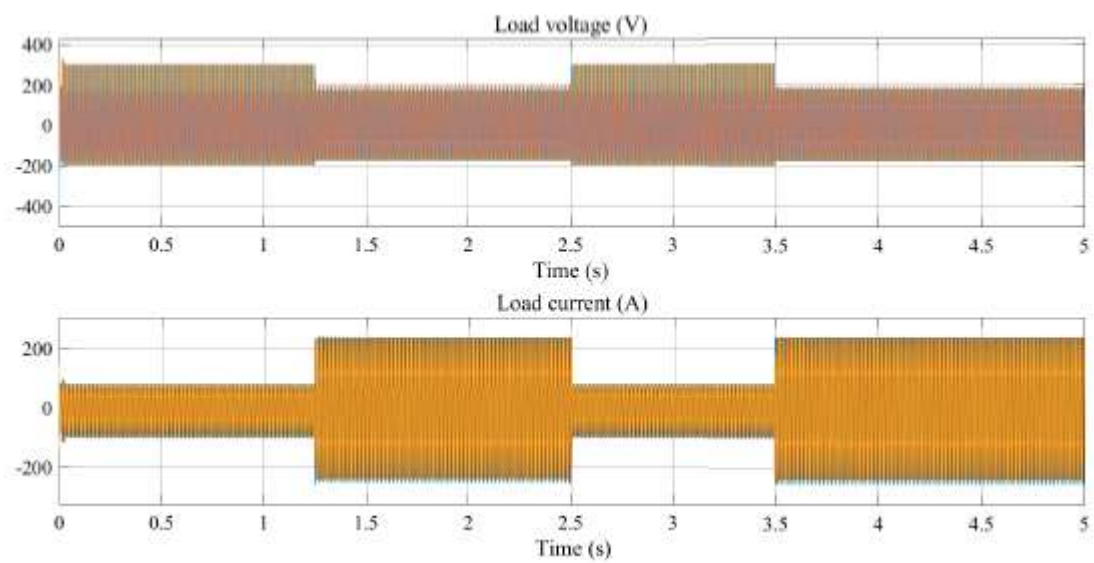
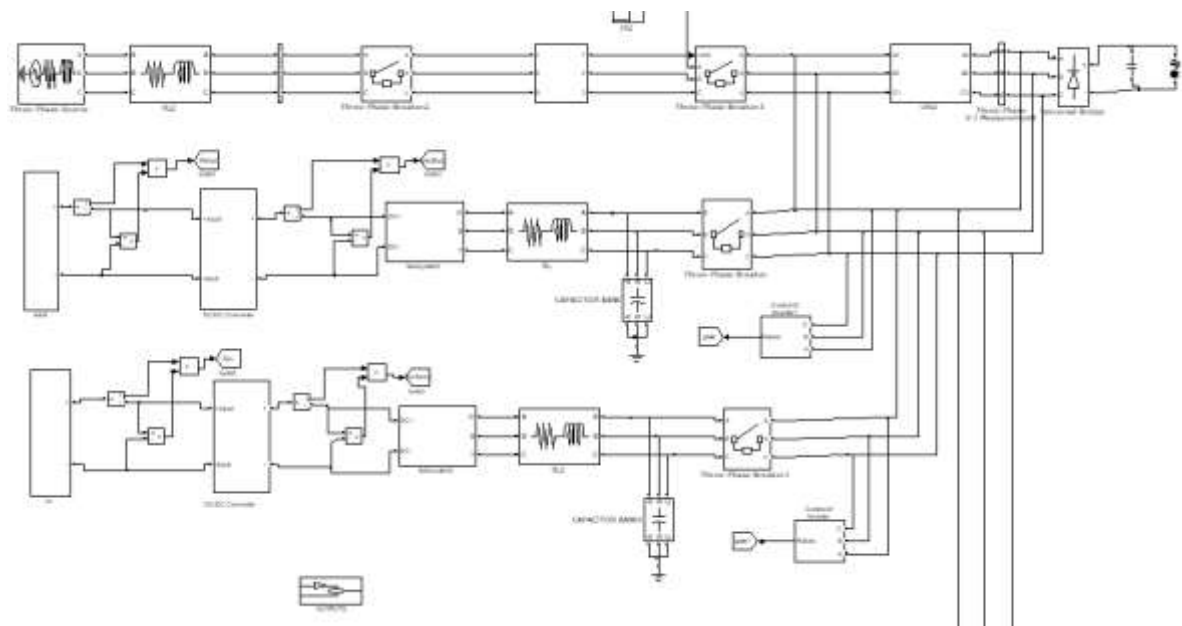
To get a reliable power transfer system and quality improvement of power, EVs play an important role in the grid connected RESs. During the availability of excess power from RESs, the charging of EVs will cost lesser which is a potential benefit for the customer, and during the deficiency of power at the grid due to increased load or unavailability of RESs, the cost of charging the EVs will be high. If EVs are used to supply power to the grid during the peak demand of the grid, it will be treated as an ESS system. The control of power flow from V2G and G2V is decided by the EVA. It also decides the number of EVs required at a particular time to charge or discharge. This makes the PV-WE-ESS-EV system reliable and improves the quality of power. The power needed to be supplied from EVs to the grid is given by equation. The number of EVs to be connected at a particular time is decided by evaluating the power required $PEVA$.

$$P_{EVA} = P_L - P_{res}$$

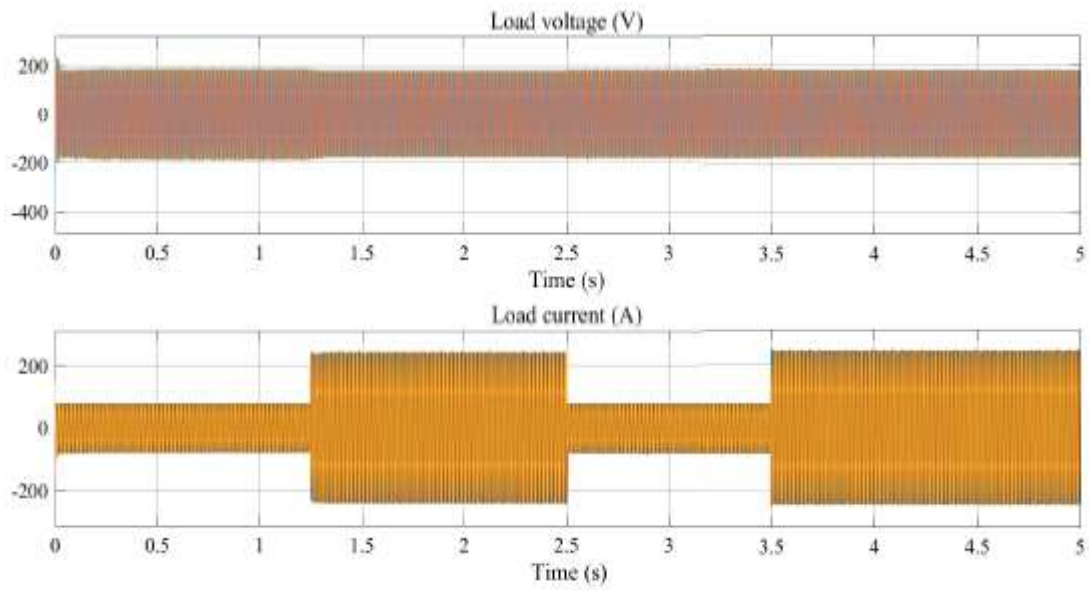
Where, $PEVA$ is the power requirement of the grid to be fulfilled by the EVs. This power is evaluated by the EVA taking the difference of load power demand, P_L , and power generated by the RESs, P_{res} . The load voltage and current are improved by providing the required power from the EVs in the case of the unavailability of PV-WE. For the validation of the proposed technique, different loading conditions are applied for a different time interval. The load voltage and current when PV-WE generation is not enough for the load demand is shown in Fig. 16a under different loading conditions. During the short period of 0 to 5 s, the PV-WE are not available in sufficient amount to fulfil the load demand, and the grid power is also less. At this condition, the voltage and current are shown clearly in Fig. 17a for the period of 0 to 2.5s. The voltage and current have distortions; therefore, THD values are high.

The qualities of voltage and current are improved on adding the EVs to the grid as shown in Fig. 16b. The excess load demand is evaluated by the EVA and the number of EVs to be connected to the grid is decided accordingly.

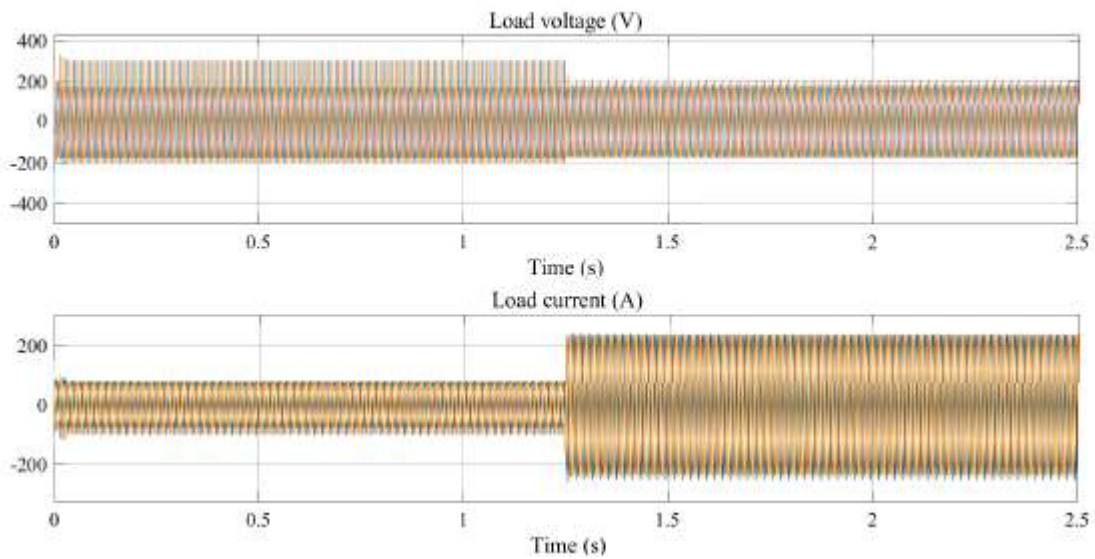
Simulation diagram



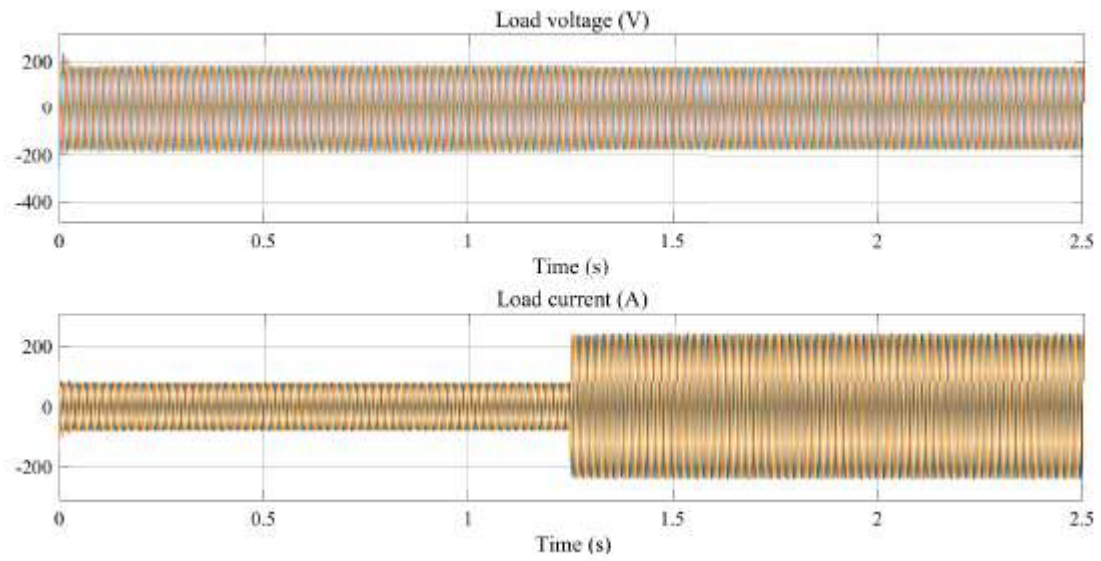
Voltage and current quality under unavailability of PV-WE and without the addition of Evs



Voltage and current quality under unavailability of PV-WE and with the addition of EVs (Waveform quality is improved).



Voltage and current of load side without EVA under the availability of PV-WE in lesser amount for time span 0 to 2.5 s shows high distortions in the different loading conditions



Voltage and current of load side with EVA under the availability of PV-WE in lesser amount for time span 0 to 2.5 s. The load power is fulfilled and the waveforms of voltage and current are improved under different loading conditions.

CONCLUSION

The authors have proposed a technique for the power enhancement of the PV-WE-ESS-EV system using FLC and UPQC controllers. The FLC-based MPPT algorithms are designed for PV and WE to extract maximum power. The UPQC provides the voltage regulation support for different loading conditions. An EVA algorithm has been developed to provide load balancing and support to the weather-dependent PV and WE systems by connecting the required number of EVs. The proposed PV-WE-ESS-EV system ensures the grid system reliability and enhances the power quality of the system because of EVs addition to the grid. It is also observed that the distortions in voltage and current waveforms are avoided using the UPQC-FLC-EVA technique. The THD of load and source-side voltages and currents are obtained and are lesser than 5% using the proposed technique, which fulfills the threshold limit of THD.

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